LDA Homework 1 (2025-2026)

Question 1

Check if Data is Balanced

From the description of the experiment we note that there is dropout during follow up. Before using any summary methods to check the mean, variance and correlation structure we first check if the data is balanced. We see that there is missingness in the data as shown in figure 1 below. Patients with higher baseline BPRS scores tend to miss more follow-up visits. There is also an increase of of the percentage of missingness over time, reaching above 40% by year 6. Because simple by time averages overweight those who remain and under represent those who dropped out (patients with higher baseline BPRS scores), they can paint a biased trend if simple bprs by time averages are used. This pattern suggests data are not missing completely at random, so we use smoothing methods that are better suited for unbalanced data

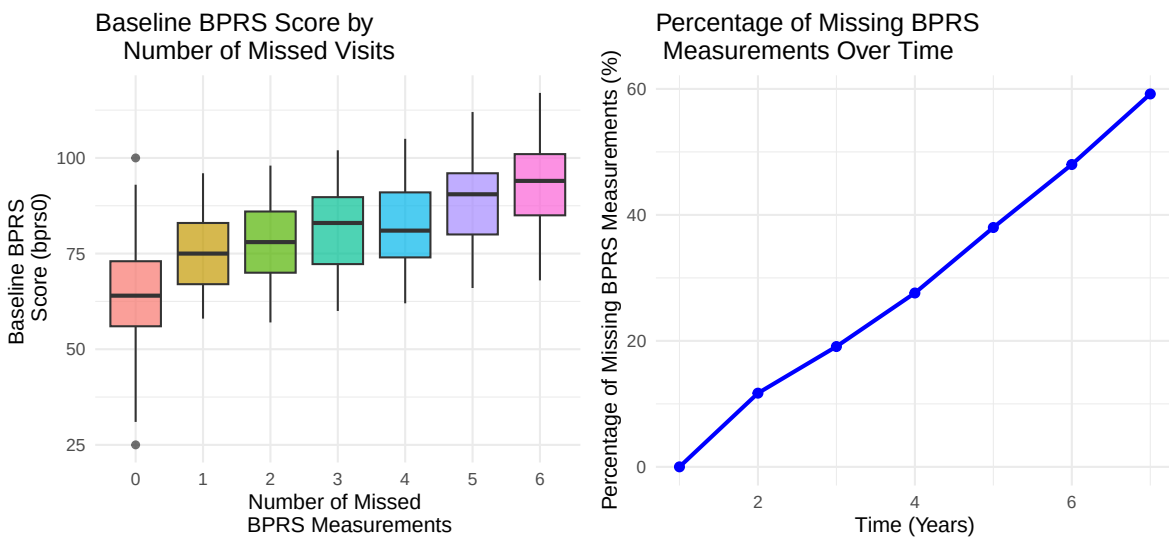


Figure 1: Baseline BPRS by number of missed visits (left) and percentage missing over time (right).

Zero-variance check for baseline Tau PET (taupet0)

From the table, taupet0 still varies through Year 2, but from Year 3 onwards it has zero variance (only one unique value among those with BPRS observed). That means taupet0 cannot explain any between-patient differences after Year 2 and coefficients for those years are not estimable. We should not use taupet0 from year 3 as this variable would not provide information on growth trajectories from here onwards.

Table 1: Baseline taupet0 variability among patients with observed BPRS at each year

n_patients	n_unique_taupet0	n_unique_abet0	sd_taupet0	sd_abet0	year	zero_variance
1253	27	67	0.1159765	0.4535830	0	FALSE
1106	13	65	0.0492840	0.4208376	1	FALSE
1014	6	62	0.0231188	0.3967379	2	FALSE
907	1	55	0.0000000	0.3730319	3	TRUE
777	1	50	0.0000000	0.3245475	4	TRUE
652	1	39	0.0000000	0.2718052	5	TRUE
511	1	22	0.0000000	0.2131188	6	TRUE

Average loess smoothing

Categorical Variables

To explore potential subgroup differences and to see how various variables are associated with the evolution of BPRS, we create LOESS smoothed plots of BPRS over time stratified by categorical variables: sex, education, job status, living situation, and clinical center. Across these variables, BPRS increases over time. Most subgroup curves are roughly parallel, showing main effects without interactions for variables like sex, job status, and living situation. The four education categories overlap almost completely and there is no difference in trend. This means there is unlikely a significant effect of education on BPRS trajectory.

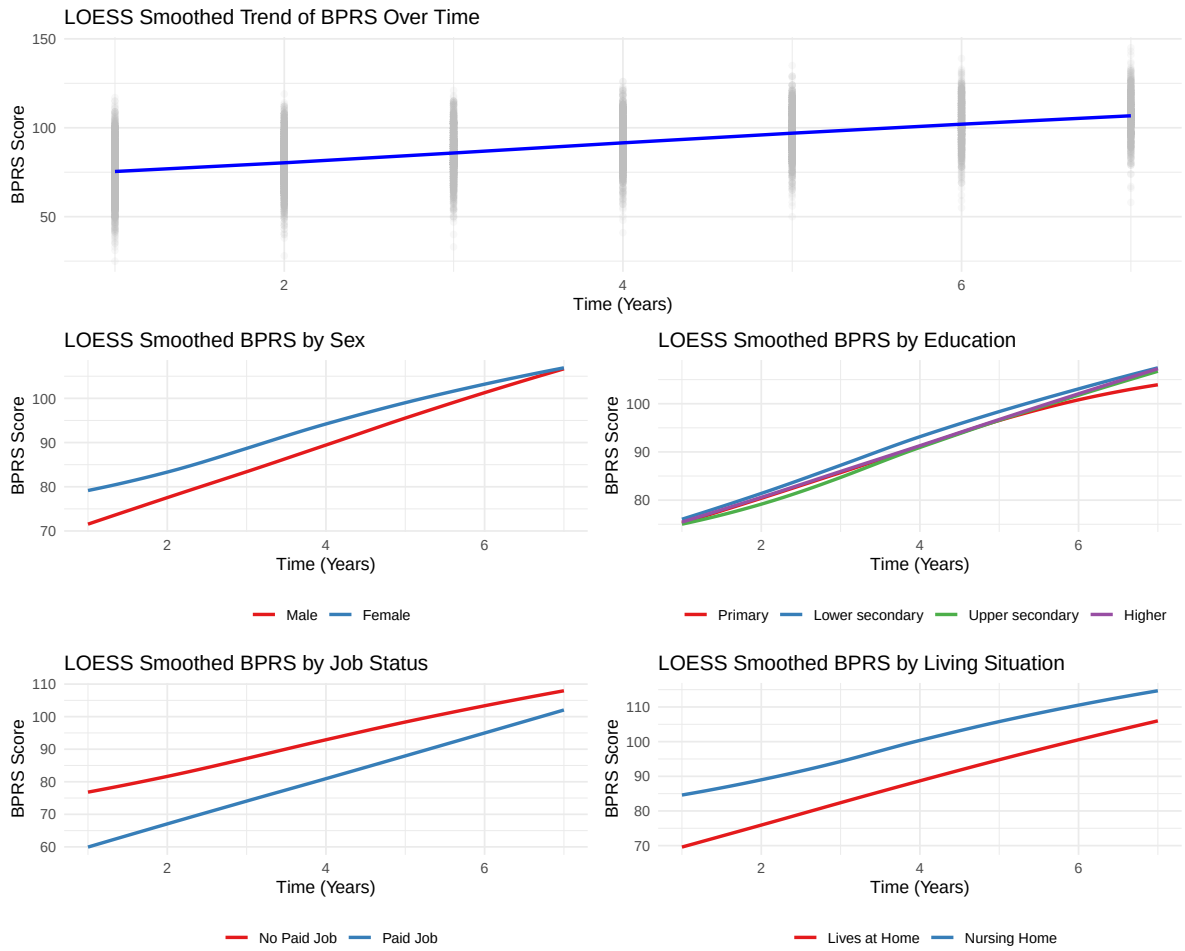


Figure 2: LOESS Smoothed BPRS Over Time by Subgroups.
Overall average (top), sex, education, job status,
living situation, and clinical center.

Continuous Variables

To explore continuous predictors, we plotted LOESS-smoothed BPRS against age, BMI and baseline CDR-SB (cdrsb0) and abept0 using separate curves for each follow-up year. Higher age is associated with higher BPRS at every visit. BMI shows a curved pattern that peaks in the mid-20s, suggesting that a non-linear term would describe it better than a straight line. Baseline CDR-SB shows only a weak and slightly curved association with BPRS. The LOESS plot of BPRS versus ABPET0 by visit year shows an unstable curved pattern, with peak at year 3 that is not repeated at other visits, which might be because of high missingness of this variable in follow up years

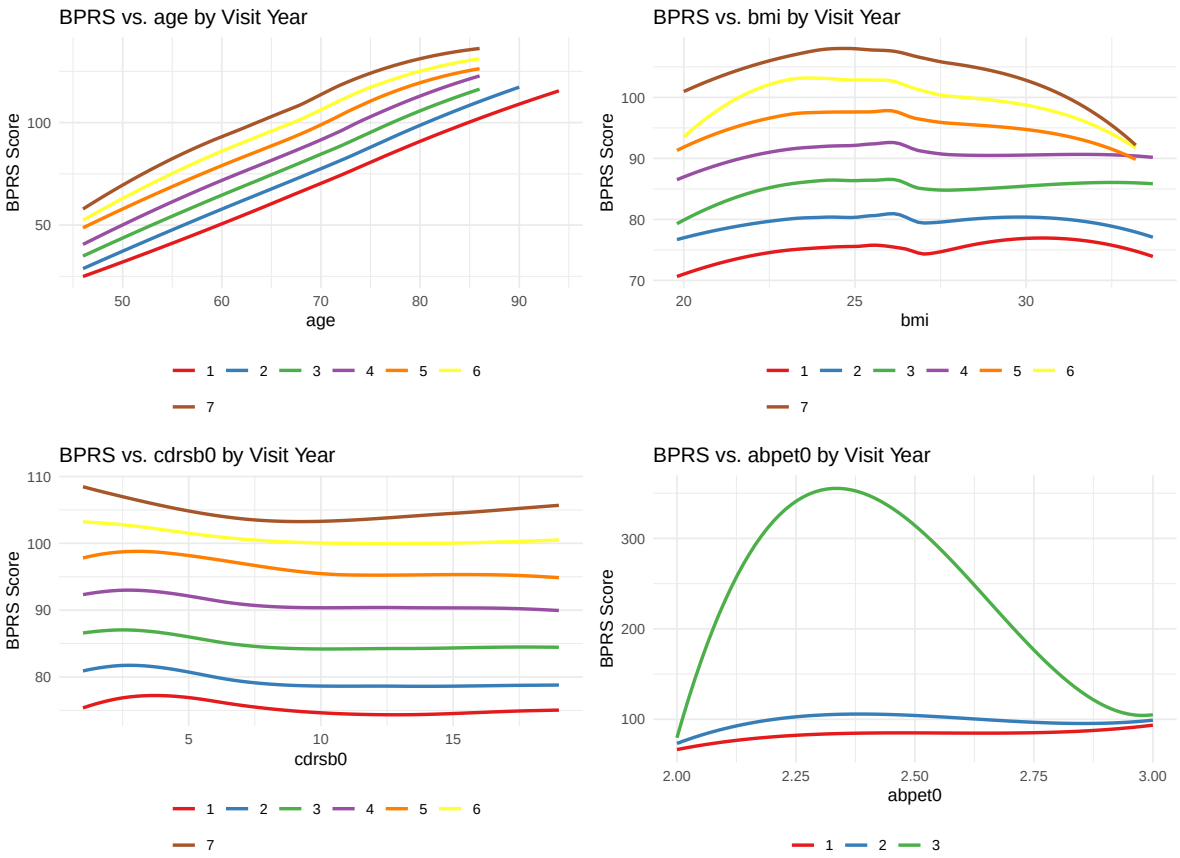


Figure 3: LOESS Smoothed BPRS Over Continuous Predictors by Visit Year.
Age, BMI, baseline BPRS (bprs0), and baseline CDR−SB (cdrsb0).

Trial Centers Plot

To check whether clinical center (trial) affects the mean structure, we plot(figure 4) LOESS-smoothed BPRS over time separately by trial centre. We see clear separate curves between centers, indicating that trial should be included as a random effect in the model to account for between center variability.

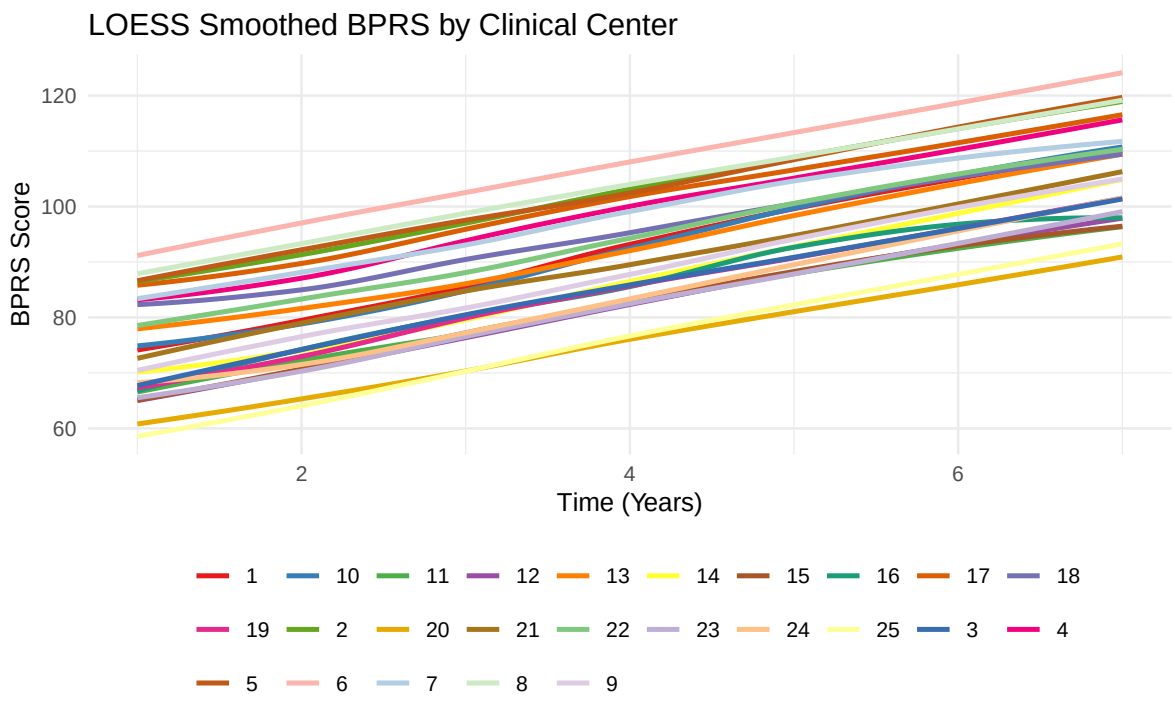


Figure 4: LOESS Smoothed BPRS Over Time by Clinical Center.

Variance & Correlation Structure

The smoothed squared residuals shows a flat residual curve, we dont observe a heteroscedastic pattern, meaning the spread of patients BPRS's readings trajectory around there expected readings doesn't change a lot. The semi-variogram also rises with the time gap between measurements, showing that observations from the same patient are more similar when close together and become less similar as the gap widens. These pattern indicates a within-patient correlation that weakens with distance in time therefore, the models we fit need to allow correlation structure that decays over time. The plots are shown in figure 5 below.

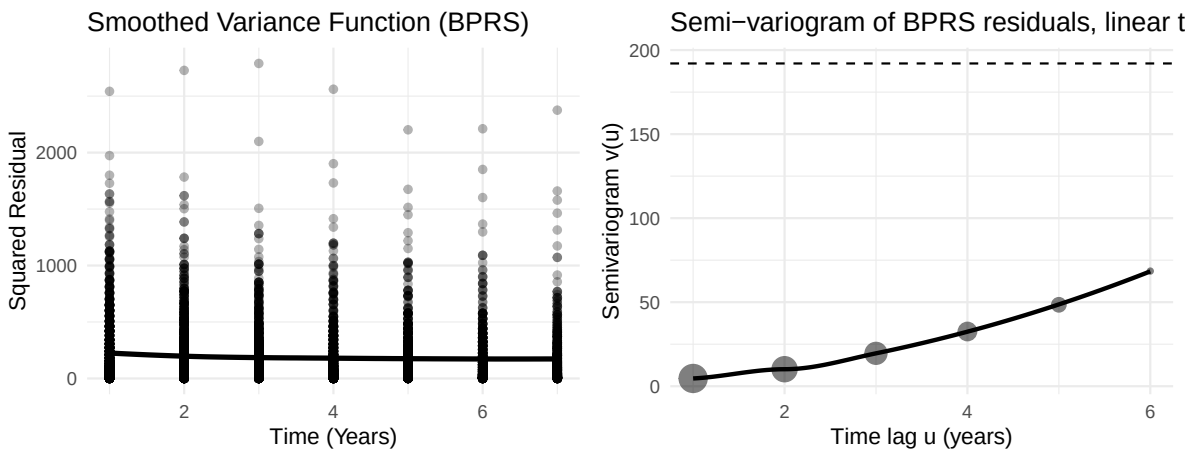


Figure 5: Smoothed Variance Function (left) and Semi−variogram (right) of BPRS Residuals.

Question 1: Conclusions and modeling implications

- Mean structure: BPRS rises steadily over time. Subgroup curves are mostly parallel with level shifts by sex, job, living situation, and clear vertical shifts by clinical center; education shows little separation.
- For continuous variables age, showed a straight line pattern while BMI showed non linear relationship with BPRS.
- Variance structure: The smoothed squared-residual function is roughly flat across visits (no visual heteroscedasticity observed). We will fit a model with un assumed heterogeneous variance and compare this with a constant variance model using AIC.
- Correlation structure: The semivariogram increases with time lag, indicating strong within-patient correlation at short lags that decays as visits are farther apart.
- Missingness: Dropout increases over time and is more common for patients with higher baseline BPRS, so simple time-point averages would be biased
- Modeling implications: For population-average inference, we will fit repeated measures model with unstructured covariance structure and compare with AR(1) correlation
- For subject-specific inference, we will fit a mixed model with patient random intercept and slope and include a random intercept for clinical center, because there is a clear clinical site differences
- we don't recommend using taupet 0 as a covariate in longitudinal models as it has zero variance from year 3 onwards, for the participants who's BPRS measurement were taken at year 3 and beyond. Hence it will not show how taupet0 affects BPRS over the years

Question 2

Individual profiles & per-patient slopes

To visualise heterogeneity in longitudinal evolution, BPRS trajectories of 40 random patients were plotted. These individual profiles show variation in baseline symptom severity, together with increasing trends over time. The profiles also illustrate the unbalanced nature of the data, as subjects contribute different numbers of observations. These observations help justify the choice of summary statistics for longitudinal data.

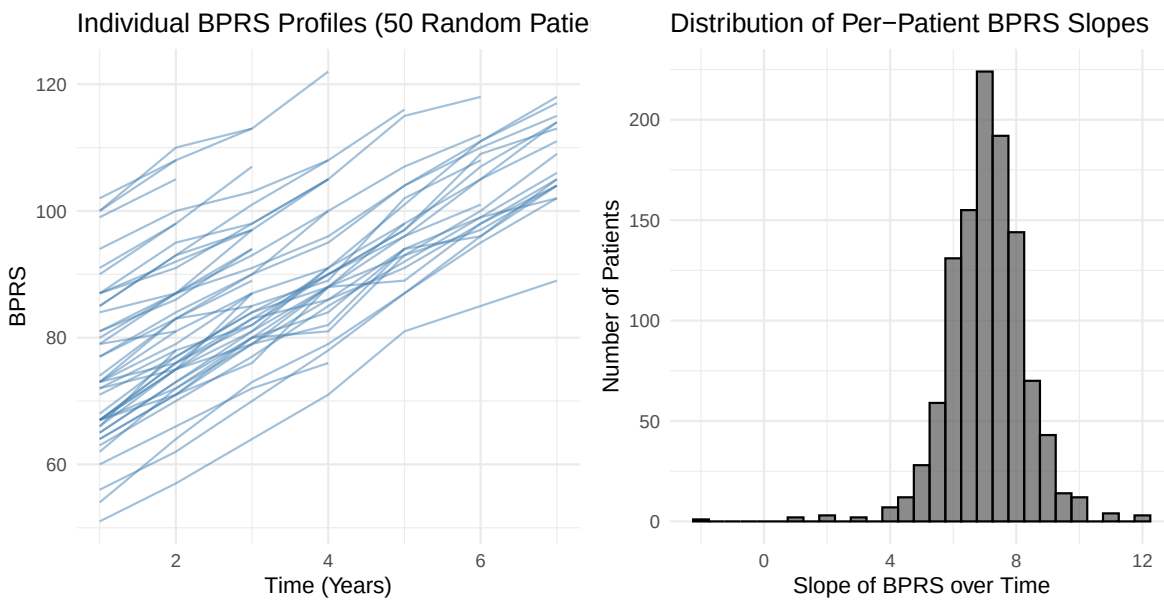


Figure 6: Individual BPRS Profiles (left) and Distribution of Per−Patient Slopes (right).

Separate analyses for each follow-up year

For each follow-up year $t \in \{1, \dots, 6\}$, we fit an ordinary linear model relating the BPRS score at year t to baseline covariates:

$$\begin{aligned} extBPRS_{it} = \alpha_{0t} + \alpha_{1t} \cdot \text{sex}_i + \alpha_{2t} \cdot \text{age}_i + \alpha_{3t} \cdot \text{bmi}_i + \alpha_{4t} \cdot \text{job}_i + \alpha_{5t} \cdot \text{adl}_i \\ + \alpha_{6t} \cdot \text{wzc}_i + \alpha_{7t} \cdot \text{cdrsb0}_i + \alpha_{8t} \cdot \text{abpet0}_i + \alpha_{9t} \cdot \text{taupet0}_i + \varepsilon_{it}. \end{aligned}$$

where, for each fixed year t:

- α_{0t} is the expected BPRS at year t for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.

- α_{1t} is the difference at year t for Female vs Male.
- α_{2t} is the change at year t per 1-year increase in age.
- α_{3t} is the change at year t per 1-unit increase in BMI.
- α_{4t} is the difference at year t for Paid Job vs No Paid Job.
- α_{5t} is the change at year t per 1-unit increase in ADL.
- α_{6t} is the difference at year t for Nursing Home vs Lives at Home.
- α_{7t} is the change at year t per 1-unit increase in baseline CDR-SB (cdrsb0).
- α_{8t} is the change at year t per 1-unit increase in baseline Amyloid PET (abpet0).
- α_{9t} is the change at year t per 1-unit increase in baseline Tau PET (taupet0).
- ε_{it} is the error term at year t.

Table 2: Separate Linear Models for Each Follow-up Year

covariate	Year0	Year1	Year2	Year3	Year4	Year5	Year6
(Intercept)	-122.72 [-126.28, -119.16]	-119.78 [-126.03, -113.52]	-105.57 [-120.15, -90.99]	-104.92 [-111.33, -98.51]	-101.46 [-109.63, -93.29]	-95.55 [-106.07, -85.02]	-87.73 [-101.67, -73.79]
abpet0	0.34 [-0.07, 0.74]	0.20 [-0.29, 0.69]	0.03 [-0.61, 0.68]	-0.13 [-0.98, 0.71]	-0.03 [-1.18, 1.13]	-0.10 [-1.74, 1.54]	-1.11 [-3.75, 1.54]
adl	2.01 [1.92, 2.11]	2.17 [2.05, 2.29]	2.25 [2.09, 2.41]	2.11 [1.91, 2.32]	2.26 [2.00, 2.52]	2.37 [2.02, 2.71]	2.42 [1.96, 2.87]
age	2.36 [2.33, 2.40]	2.41 [2.36, 2.45]	2.42 [2.37, 2.48]	2.41 [2.34, 2.48]	2.45 [2.36, 2.54]	2.48 [2.37, 2.58]	2.50 [2.37, 2.64]
bmi	0.49 [0.44, 0.55]	0.52 [0.45, 0.58]	0.51 [0.42, 0.59]	0.54 [0.43, 0.65]	0.56 [0.42, 0.70]	0.51 [0.33, 0.69]	0.47 [0.23, 0.71]
cdrsb0	-0.02 [-0.04, -0.01]	-0.04 [-0.06, -0.02]	-0.06 [-0.08, -0.04]	-0.08 [-0.11, -0.05]	-0.13 [-0.17, -0.09]	-0.15 [-0.20, -0.10]	-0.17 [-0.23, -0.10]
jobPaid Job	-18.33 [-19.09, -17.57]	-19.24 [-20.17, -18.30]	-19.77 [-20.97, -18.56]	-19.01 [-20.54, -17.47]	-20.19 [-22.13, -18.25]	-20.86 [-23.30, -18.41]	-21.07 [-24.18, -17.96]
sexFemale	-0.28 [-0.51, -0.05]	-0.40 [-0.68, -0.12]	-0.18 [-0.54, 0.18]	-0.22 [-0.68, 0.25]	-0.34 [-0.94, 0.26]	-0.37 [-1.15, 0.41]	-0.57 [-1.64, 0.49]
taupet0	0.43 [-0.57, 1.43]	0.37 [-2.34, 3.09]	-4.06 [-11.47, 3.35]	NA	NA	NA	NA
wzcNursing Home	1.69 [1.43, 1.94]	1.94 [1.63, 2.25]	2.10 [1.69, 2.51]	2.24 [1.69, 2.80]	1.94 [1.18, 2.71]	1.43 [0.34, 2.52]	2.16 [0.42, 3.90]

Question 3

Motivation Repeated-Measures Model

To estimate population effects, we use a repeated-measures multivariate model. This approach treats each patient's yearly BPRS readings dependent. A normal multivariate linear model assumes all observations are independent, which is not realistic here because the same patient BPRS measurements are taken yearly up to year 6. The repeated measures model correctly handles the correlation within a person, the uneven follow-up due to dropout, and the changing variability across visits.This model uses all available data and gives more reliable averages and standard errors. We start from a saturated repeated measures model where time, all baseline characteristics, and their interactions with time are included. This gives us a way to assess how the variable affect both the starting BPRS level and the change over time, so that we do not miss important effects when we later simplify the model. For the repeated measurements within a patient, we allow a very flexible pattern for how BPRS scores at different visits are correlated to each other, without forcing a specific correlation pattern in advance. We do not use this saturated model to report final clinical results, we use it as a screening model and then simplify the mean structure.

In question 2 we have yearly results(Table 2) but analysing each year separately would mean

running several models for the same question/hypothesis. This increases the chance of getting a significant result just by chance (Type 1 error), because we are not using all observations together it also leads to lower power. Separate analyses also do not answer the main question of the study, which is how BPRS changes over time and how baseline factors relate to that change. They ignore that measurements from the same patient are correlated, which can give misleading results. The repeated-measures model avoids these problems by using all time points at once and by correctly accounting for the fact that each patient contributes several, correlated measurements.

$$
\begin{aligned}
E[Y_{ij}|X_i] = & \beta_0 + \beta_1 \cdot \text{time}_j + \beta_2 \cdot \text{edu}_i + \beta_3 \cdot \text{sex}_i + \beta_4 \cdot \text{age}_i + \beta_5 \cdot \text{bmi}_i + \beta_6 \cdot \text{job}_i \\
& + \beta_7 \cdot \text{adl}_i + \beta_8 \cdot \text{wzc}_i + \beta_9 \cdot \text{cdrsb0}_i + \beta_{10} \cdot \text{abpet0}_i \\
& + \beta_{11} \cdot (\text{time}_j \times \text{sex}_i) + \beta_{12} \cdot (\text{time}_j \times \text{age}_i) + \beta_{13} \cdot (\text{time}_j \times \text{bmi}_i) \\
& + \beta_{14} \cdot (\text{time}_j \times \text{job}_i) + \beta_{15} \cdot (\text{time}_j \times \text{adl}_i) + \beta_{16} \cdot (\text{time}_j \times \text{wzc}_i) \\
& + \beta_{17} \cdot (\text{time}_j \times \text{cdrsb0}_i) + \beta_{18} \cdot (\text{time}_j \times \text{abpet0}_i)
\end{aligned}
$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the intercept: expected BPRS at time 0 for the reference patient (Male, Primary education, No Paid Job, Lives at Home), with continuous covariates at 0.
- β_1 is the main effect of time: average yearly change in BPRS for the reference patient.
- β_2 is the (vector of) main effect(s) of education: contrasts for Lower secondary, Upper secondary, and Higher vs Primary (written compactly as $\beta_2 \, \text{edu}_i$).
- β_3 is the main effect of sex: Female vs Male at time 0.
- β_4 is the main effect of age: per 1-year increase at baseline.
- β_5 is the main effect of BMI: per 1-unit increase at baseline.
- β_6 is the main effect of job: Paid Job vs No Paid Job at time 0.
- β_7 is the main effect of ADL: per 1-unit increase at baseline.
- β_8 is the main effect of living situation (wzc): Nursing Home vs Lives at Home at time 0.
- β_9 is the main effect of baseline CDR-SB (cdrsb0): per 1-unit increase at baseline.
- β_{10} is the main effect of baseline Amyloid PET (abpet0): per 1-unit increase at baseline.
- β_{11} is the time-by-sex interaction: additional yearly change for Female vs Male.
- β_{12} is the time-by-age interaction: modification of yearly change per 1-year increase in age.
- β_{13} is the time-by-BMI interaction: modification of yearly change per 1-unit increase in BMI.
- β_{14} is the time-by-job interaction: additional yearly change for Paid Job vs No Paid Job.

- β_{15} is the time-by-ADL interaction: modification of yearly change per 1-unit increase in ADL.
- β_{16} is the time-by-living situation (wzc) interaction: additional yearly change for Nursing Home vs Lives at Home.
- β_{17} is the time-by-cdrsb0 interaction: modification of yearly change per 1-unit increase in baseline CDR-SB.
- β_{18} is the time-by-abpet0 interaction: modification of yearly change per 1-unit increase in baseline Amyloid PET.

Notes: Categorical predictors (sex, edu, job, wzc) are encoded as factors in R. Their coefficients represent contrasts relative to the reference levels shown above; for education, the single symbol β_2 denotes a vector of three contrasts (levels 2–4 versus Primary).

The within-patient errors $e_i = (e_{i1}, \dots, e_{in_i})^T$ are assumed to follow a multivariate normal distribution, $e_i \sim N(0, \Sigma_i)$, where Σ_i is a covariance matrix that accounts for correlation and heteroscedasticity. In this model, we use an unstructured correlation matrix (`corSymm`) and allow for different variances at each time point (`varIdent`).

Note: taupet0 terms (main effect and time interaction) were intentionally excluded from this model due to near-zero variability at later follow-up waves (see earlier diagnostic section), which prevents stable estimation of their effects in a marginal mean model.

The complex repeated-measures model that includes time, all baseline characteristics, and all their interactions with time, while allowing for unstructured covariance structure. This complex model confirms that BPRS increases over time and that age, BMI, ADL, job status, and living in a nursing home are all associated to BPRS levels. However, most of interactions between time and other covariates are not significant, which tells us we can safely move to a simpler mean structure without losing important clinical information.

Table 3: Gls Full Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-129.58 [-133.00, -126.15]	1.75	-74.20	<0.01
time	5.96 [4.27, 7.64]	0.86	6.91	<0.01
eduLower secondary	0.20 [-0.13, 0.52]	0.17	1.19	0.23
eduUpper secondary	0.12 [-0.19, 0.42]	0.16	0.75	0.46
eduHigher	0.19 [-0.11, 0.50]	0.16	1.23	0.22
sexFemale	-0.24 [-0.49, 0.00]	0.13	-1.95	0.05
age	2.36 [2.32, 2.40]	0.02	126.37	<0.01
bmi	0.51 [0.45, 0.57]	0.03	17.00	<0.01
jobPaid Job	-18.13 [-18.91, -17.35]	0.40	-45.69	<0.01
adl	2.00 [1.90, 2.10]	0.05	38.57	<0.01
wzcNursing Home	1.63 [1.35, 1.92]	0.15	11.16	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.21	0.83
abpet0	0.34 [-0.10, 0.78]	0.22	1.52	0.13
time:sexFemale	-0.03 [-0.16, 0.09]	0.06	-0.52	0.61
time:age	0.01 [-0.00, 0.03]	0.01	1.59	0.11
time:bmi	-0.00 [-0.03, 0.03]	0.01	-0.06	0.95
time:jobPaid Job	-0.39 [-0.78, 0.01]	0.20	-1.90	0.06
time:adl	0.05 [-0.00, 0.11]	0.03	1.95	0.05
time:wzcNursing Home	0.14 [-0.00, 0.29]	0.07	1.90	0.06
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.10	<0.01
time:abpet0	-0.07 [-0.29, 0.16]	0.11	-0.58	0.56

Simplified Mean Structure

For this step we fit a repeated-measures model with a simplified mean structure but the same unstructured covariance as before. In this model, time and all main baseline characteristics except education remain in the mean structure, and only one time interaction (time with baseline CDR-SB) is kept. The estimated correlation matrix shows a positive correlation between repeated BPRS scores within a patient, especially for visits that are closer in time, and the visit-specific standard deviations increase from baseline to later years, contradicting with our earlier variance plots showing homoscedastic variance. We will use this reduced-mean, unstructured covariance model as the starting point when we compare with simpler covariance structures for instance AR(1)

$$\begin{aligned} E[Y_{ij}|X_i] =& \beta_0 + \beta_1 \cdot \text{time}_j + \beta_2 \cdot \text{sex}_i + \beta_3 \cdot \text{age}_i + \beta_4 \cdot \text{bmi}_i + \beta_5 \cdot \text{job}_i \\ &+ \beta_6 \cdot \text{adl}_i + \beta_7 \cdot \text{wzc}_i + \beta_8 \cdot \text{cdrsb0}_i + \beta_9 \cdot \text{taupet0}_i + \beta_{10} \cdot \text{abpet0}_i \\ &+ \beta_{11} \cdot (\text{time}_j \times \text{cdrsb0}_i) \end{aligned}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the intercept: expected BPRS at time 0 for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- β_1 is the main effect of time: average yearly change in BPRS for the reference patient.
- β_2 is the main effect of sex: Female vs Male at time 0.
- β_3 is the main effect of age: per 1-year increase at baseline.
- β_4 is the main effect of BMI: per 1-unit increase at baseline.
- β_5 is the main effect of job: Paid Job vs No Paid Job at time 0.
- β_6 is the main effect of ADL: per 1-unit increase at baseline.
- β_7 is the main effect of living situation (wzc): Nursing Home vs Lives at Home at time 0.
- β_8 is the main effect of baseline CDR-SB (cdrsb0): per 1-unit increase at baseline.
- β_9 is the main effect of baseline Tau PET (taupet0): per 1-unit increase at baseline.
- β_{10} is the main effect of baseline Amyloid PET (abpet0): per 1-unit increase at baseline.
- β_{11} is the time-by-cdrsb0 interaction: modification of yearly change per 1-unit increase in baseline CDR-SB.

The covariance structure remains the same as in the full model.

Table 4: Simplfied mean structiure Unstructured Covariance Results

term	estimate	std.error	statistic	p.value
(Intercept)	-131.59 [-134.11, -129.08]	1.28	-102.43	<0.01
time	7.22 [7.14, 7.30]	0.04	179.38	<0.01
sexFemale	-0.27 [-0.46, -0.07]	0.10	-2.72	<0.01
age	2.39 [2.37, 2.42]	0.01	212.64	<0.01
bmi	0.51 [0.47, 0.55]	0.02	22.80	<0.01
jobPaid Job	-18.69 [-19.30, -18.08]	0.31	-59.80	<0.01
adl	2.08 [2.01, 2.16]	0.04	53.17	<0.01
wzcNursing Home	1.84 [1.62, 2.05]	0.11	16.86	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.35	0.73
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.29	<0.01

LR test between full and reduced mean structure models

The likelihood ratio test compares the complex mean structure model (with all time–covariate interactions) to the reduced model (only time × CDR-SB). The test statistic is 13.78 with a p-value of 0.25, which means there is no evidence that the extra interaction terms improve the fit. This shows the simpler mean structure explains the data just as well as the saturated one, while using fewer parameters, so we will keep the reduced model as our working mean structure.

Table 5: Likelihood Ratio Test between Full and Reduced Mean Structure Models

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
gls_full_mean_structure	1	49	27318.29	27648.33	-13610.14		NA	NA
gls_reduced_mean_structure	2	38	27310.07	27566.02	-13617.03	1 vs 2	13.78	0.25

Reduced Covariance Structure

For this step, we keep the same simplified mean structure and only change how we model the correlation over time. The aim is to check whether a simpler covariance structure is sufficient. We fit a repeated-measures model with an AR(1) correlation structure, where each visit is highly correlated with the previous one and the correlation weakens as the time gap increases. We still allow variability to differ by visit. This means BPRS measurements taken close in time tend to be similar, and later visits show more spread.

When we compare this AR(1) model to the more flexible model with a unstructured covariance structure, the likelihood ratio test shows (L.Ratio ≈ 1233, $p < 0.0001$) and AIC/BIC worsen for AR(1). This indicates that the data pattern is too complex to be captured by this AR(1)

correlation structure, so for population-average inference we keep the unstructured covariance from the previous model.

Table 6: Marginal Model with AR(1) Covariance Results

term	estimate	std.error	statistic	p.value
(Intercept)	-131.98 [-135.20, -128.75]	1.64	-80.27	<0.01
time	7.24 [7.17, 7.31]	0.03	208.57	<0.01
sexFemale	-0.28 [-0.53, -0.03]	0.13	-2.24	0.03
age	2.40 [2.37, 2.43]	0.01	166.48	<0.01
bmi	0.50 [0.45, 0.56]	0.03	17.57	<0.01
jobPaid Job	-18.91 [-19.69, -18.12]	0.40	-47.07	<0.01
adl	2.10 [2.00, 2.20]	0.05	41.83	<0.01
wzcNursing Home	1.84 [1.57, 2.12]	0.14	13.25	<0.01
cdrsb0	0.00 [-0.02, 0.03]	0.01	0.33	0.74
time:cdrsb0	-0.02 [-0.03, -0.02]	0.00	-6.67	<0.01

Reduced mean structure vs reduced covariance structure

Table 7: Likelihood Ratio Test between Reduced Mean Structure and Reduced Covariance Structure Models

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
gls_reduced_mean_structure	1	38	27310.07	27566.02	-13617.03		NA	NA
gls_reduced_cov_structure	2	18	28503.30	28624.54	-14233.65	1 vs 2	1233.23	0

Discussion of Question 3 Results

For Question 3, our final repeated-measures model uses a reduced mean structure with a unstructured covariance. On the mean side, BPRS increases on average by about 7 points per year. After adjustment, higher age, higher BMI, worse ADL, job status, and living in a nursing home are all associated with higher BPRS levels, while women have slightly lower BPRS than men. The interaction of time and CDR-SB term is is significant meaning patients who start with better cognition tend to show a faster increase in BPRS, whereas those who already have worse cognition at baseline worsen more slowly. This directly answers our main question at the population level: BPRS worsens over time, and the speed of worsening is associated with baseline cognitive status.

For the biomarkers, we focus on ABPET0 and CDR-SB. ABPET0 is retained in the marginal model, but its coefficient is small and the confidence interval is wide, so its association with BPRS is uncertain. The LOESS plot of BPRS versus ABPET0 by visit year shows an unstable

curved pattern, with a peak at year 3 that is not repeated at other visits. This could be due to data missingness. TAUPET0, on the other hand, was dropped because from year 3 onwards it has no variability, so it cannot explain differences in BPRS.

The covariance part of the model lets each visit have its own variability and allows all time points within a patient to be correlated without imposing a correlation structure. This matches what we saw in the exploratory plots repeated BPRS measurements from the same patient are strongly correlated, especially when visits are close in time, and the spread increases over follow-up. Using this unstructured structure gives more realistic standard errors for the population averages.

Finally, this marginal model has some limitations compared with a mixed-effects model. It provides population-average effects but not patient-specific trajectories, and it does not explicitly model differences between clinical centres. The LOESS plot by centre shows centres with roughly parallel but clearly non overlapping curves for majority of centres, suggesting centre level differences in BPRS levels over time. In the current model, this extra variation is absorbed into the residual covariance rather than being separated out. In Question 5 we therefore move to a mixed-effects model with random intercepts and slopes for patients, and a random intercept for trial, to better describe individual and centre-specific disease trajectories while still addressing the main objective of how BPRS changes over time, and how those changes relate to baseline clinical characteristics, cognition, and biomarkers.

Question 4

In the first stage of a two-stage analysis, we fit a separate linear regression model for each patient to estimate their individual intercept and slope over time. For each patient i, the model is:

$$Y_{ij} = \beta_{0i} + \beta_{1i} \cdot \text{time}_j + e_{ij}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_{0i} is the patient-specific intercept (estimated baseline BPRS).
- β_{1i} is the patient-specific slope (estimated rate of change in BPRS).
- e_{ij} are the individual error terms, assumed to be independent with mean zero.

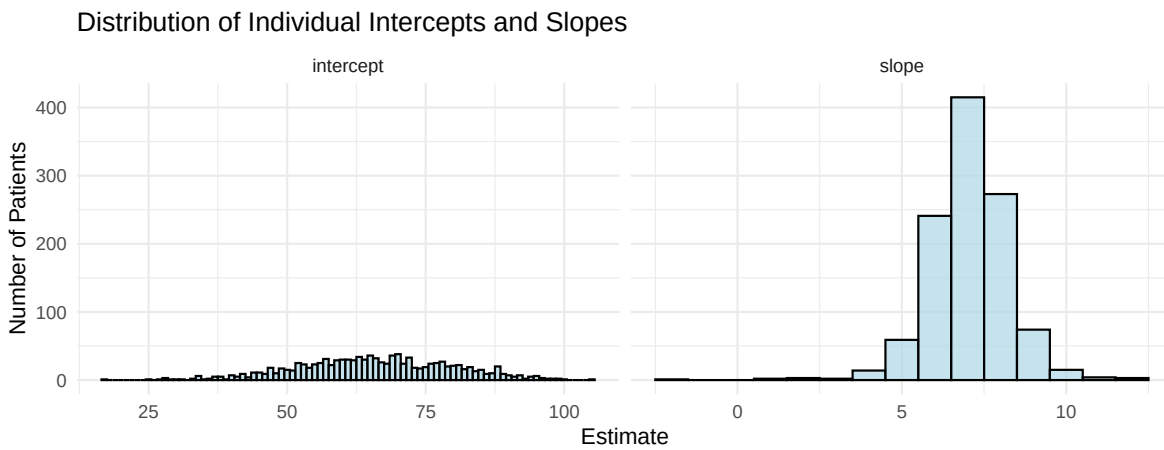


Figure 7: Individual BPRS Profiles (left) and
Distribution of Individual Intercepts and Slopes (right).

Stage 2: Relate Intercepts and Slopes to Covariates

Model for the Intercepts

In the second stage, we model the estimated patient-specific intercepts ($\hat{\beta}_{0i}$) as a function of baseline covariates to understand which factors predict the initial BPRS score. The model is:

$$\begin{aligned}\hat{\beta}_{0i} =& \gamma_{00} + \gamma_{01} \cdot \text{sex}_i + \gamma_{02} \cdot \text{age}_i + \gamma_{03} \cdot \text{bmi}_i + \gamma_{04} \cdot \text{job}_i + \gamma_{05} \cdot \text{adl}_i \\ &+ \gamma_{06} \cdot \text{wzc}_i + \gamma_{07} \cdot \text{cdrsb0}_i + \gamma_{08} \cdot \text{abpet0}_i + \gamma_{09} \cdot \text{taupet0}_i + \epsilon_{0i}\end{aligned}$$

where:

- $\hat{\beta}_{0i}$ is the estimated intercept for patient i from Stage 1.
- γ_{00} is the average intercept for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{01} is the intercept difference for Female vs Male.
- γ_{02} is the intercept change per 1-year increase in age.
- γ_{03} is the intercept change per 1-unit increase in BMI.
- γ_{04} is the intercept difference for Paid Job vs No Paid Job.
- γ_{05} is the intercept change per 1-unit increase in ADL.
- γ_{06} is the intercept difference for Nursing Home vs Lives at Home.
- γ_{07} is the intercept change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{08} is the intercept change per 1-unit increase in baseline Amyloid PET (abpet0).

- γ_{09} is the intercept change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{0i} is the error term.

Table 8: Two-Stage Intercept Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-130.85 [-137.31, -124.39]	3.29	-39.76	<0.01
sexFemale	-0.22 [-0.51, 0.06]	0.15	-1.52	0.13
age	2.36 [2.32, 2.41]	0.02	101.88	<0.01
bmi	0.51 [0.45, 0.58]	0.03	15.07	<0.01
jobPaid Job	-18.28 [-19.25, -17.32]	0.49	-37.15	<0.01
adl	2.03 [1.90, 2.15]	0.06	31.98	<0.01
wzcNursing Home	1.54 [1.22, 1.86]	0.16	9.41	<0.01
cdrsb0	0.00 [-0.02, 0.02]	0.01	0.12	0.90
abpet0	0.35 [-0.16, 0.86]	0.26	1.35	0.18
taupet0	0.53 [-2.27, 3.33]	1.43	0.37	0.71

Model for the Slopes

Similarly, we model the estimated patient-specific slopes ($\hat{\beta}_{1i}$) as a function of baseline covariates to see what predicts the rate of change in BPRS. The model is:

$$\begin{aligned}\hat{\beta}_{1i} =& \gamma_{10} + \gamma_{11} \cdot \text{sex}_i + \gamma_{12} \cdot \text{age}_i + \gamma_{13} \cdot \text{bmi}_i + \gamma_{14} \cdot \text{job}_i + \gamma_{15} \cdot \text{adl}_i \\ &+ \gamma_{16} \cdot \text{wzc}_i + \gamma_{17} \cdot \text{cdrsb0}_i + \gamma_{18} \cdot \text{abpet0}_i + \gamma_{19} \cdot \text{taupet0}_i + \epsilon_{1i}\end{aligned}$$

where:

- $\hat{\beta}_{1i}$ is the estimated slope for patient i from Stage 1.
- γ_{10} is the average slope for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{11} is the slope difference for Female vs Male.
- γ_{12} is the slope change per 1-year increase in age.
- γ_{13} is the slope change per 1-unit increase in BMI.
- γ_{14} is the slope difference for Paid Job vs No Paid Job.
- γ_{15} is the slope change per 1-unit increase in ADL.
- γ_{16} is the slope difference for Nursing Home vs Lives at Home.
- γ_{17} is the slope change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{18} is the slope change per 1-unit increase in baseline Amyloid PET (abpet0).

- γ_{19} is the slope change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{1i} is the error term.

Table 9: Two-Stage Slope Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	6.37 [2.97, 9.77]	1.73	3.68	<0.01
sexFemale	-0.04 [-0.19, 0.11]	0.08	-0.55	0.58
age	0.01 [-0.01, 0.04]	0.01	1.02	0.31
bmi	0.00 [-0.03, 0.04]	0.02	0.09	0.93
jobPaid Job	-0.34 [-0.85, 0.17]	0.26	-1.32	0.19
adl	0.05 [-0.02, 0.11]	0.03	1.37	0.17
wzcNursing Home	0.19 [0.02, 0.36]	0.09	2.23	0.03
cdrsb0	-0.02 [-0.03, -0.01]	0.01	-4.30	<0.01
abpet0	-0.07 [-0.34, 0.20]	0.14	-0.53	0.60
taupet0	-0.13 [-1.61, 1.34]	0.75	-0.17	0.86

Question 5

Plausible random effects model

To understand how psychiatric symptoms evolve over time in Alzheimer's disease, we developed a statistical model that accounts for repeated measurements in each patient. This model allows each patient to have their own starting level of symptoms and their own rate of change over time. The outcome of interest was the Brief Psychiatric Rating Scale (BPRS), measured annually over six years.

We model the longitudinal BPRS scores using a linear mixed-effects model with random effects at both the patient and trial level:

$$\text{BPRS}_{it} = (\beta_0+b_{0i}+u_{0j})+(\beta_1+b_{1i})\,\text{time}_{it}+\beta_2\text{sex}_i+\beta_3\text{age}_i+\beta_4\text{bmi}_i+\beta_5\text{job}_i+\beta_6\text{adl}_i+\beta_7\text{wzc}_i+\beta_8\text{cdrsb0}_i+\beta_9(\text{time}_i$$

where:

- (BPRS_{it}) is the BPRS score for patient (i) in trial (j) at time (t)
- (_0, ..., _9) are **fixed (population-level) effects**

- (b_{0i}) is the **patient-specific random intercept**
- (b_{1i}) is the **patient-specific random slope** for time
- (u_{0j}) is the **trial-specific random intercept**
- (_{it}) is the **residual error term**

The patient-level random effects are assumed to follow a bivariate normal distribution:

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{b0}^2 & \sigma_{b0,b1} \\ \sigma_{b0,b1} & \sigma_{b1}^2 \end{pmatrix} \right),$$

and the trial-level random intercept follows:

$$u_{0j} \sim N(0, \sigma_{\text{trial}}^2).$$

Model Motivation and Justification

Here, We chose to fit linear mixed effects model because the dataset contains repeated BPRS measurements for the same patients over 6 years and across multiple clinical trials, with clear variability both in baseline symptom severity and rate of change between patients. This model was appropriate because it:

1) Explicitly accounts for correlation within patients (repeated measures from the same individual),
2) Allows patient-specific trajectories (subject-specific intercepts and slopes),
3) Uses all available data, including patients with incomplete follow-up.
4) and Adjusts the individual trajectories for baseline patient characteristics, improving precision.

Choice of Technique

An unstructured covariance matrix was used for the random effects, allowing the intercept and slope to be correlated. The model included:

1) Random intercepts and random slopes for each patient, reflecting that individuals differ both in their baseline psychiatric symptom severity and their rate of change over time.

2) A random intercept for trial was added to account for systematic differences across the 25 clinical trials.
3) Fixed effects included all clinically relevant baseline covariates (age, sex, BMI, job status, ADL, living situation, CDR-SB, amyloid PET), ensuring adjustment for major patient characteristics.
4) An interaction term between time $\times$ CDRSB was included to assess whether cognitive impairment at baseline affects the rate of psychiatric symptom progression.

Estimation Method

The model was fitted using Restricted Maximum Likelihood (REML), which provides unbiased and stable estimates of variance components, especially because our dataset had many clusters (patients, trials).We also preferred this (REML)for comparing models that differ in their random-effects structure.

Assumptions made

1) Linearity: BPRS changes linearly over the follow-up period.
2) Normality: Random effects and residual errors are approximately normally distributed.
3) Independence: After accounting for random effects, residuals are independent.
4) Missing data (dropout) are at least missing at random (MAR) conditional on the covariates and random effects.
5) Correct Covariance Structure: The random intercept–slope covariance captures individual variability adequately

Advantages

1) Properly accounts for correlation between repeated measures within patients.
2) Handles unbalanced data, allowing inclusion of patients with incomplete follow-up.
3) Produces subject-specific predictions (intercepts and slopes).
4) The model incorporates trial-level variability, improving generalizability.
5) More efficient and less biased than two-stage or marginal models for this scientific question.

Disadvantages / Potential Problems

1) More complex to interpret for non-statisticians.
2) Results depend on correct specification of random-effects structure.
3) Convergence issues can arise with complex models.

4) Assumptions such as normality of random effects and linearity may not fully hold in real data.

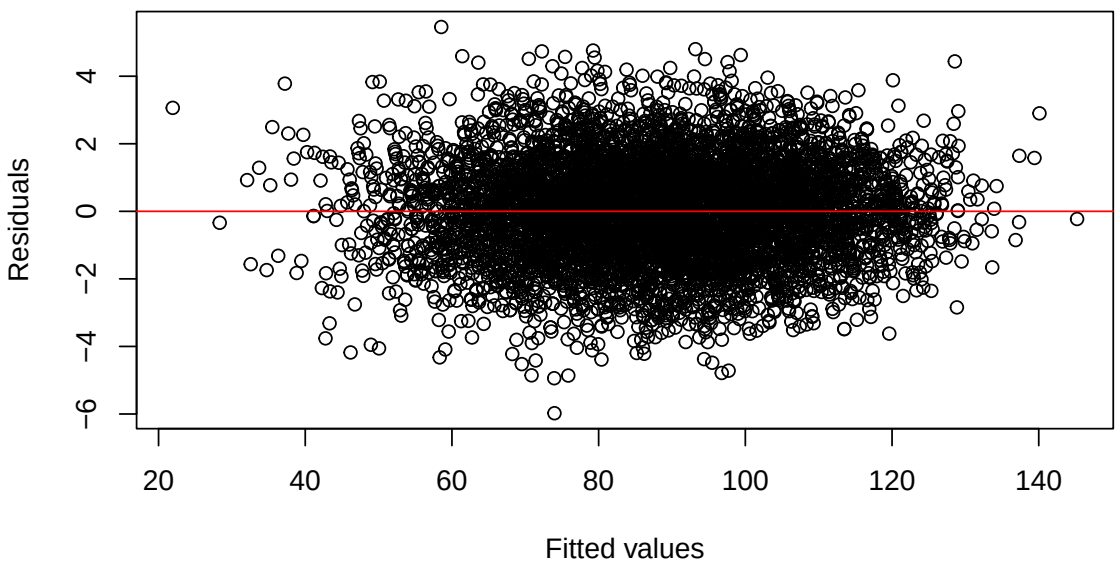
5) MAR assumption cannot be fully verified.

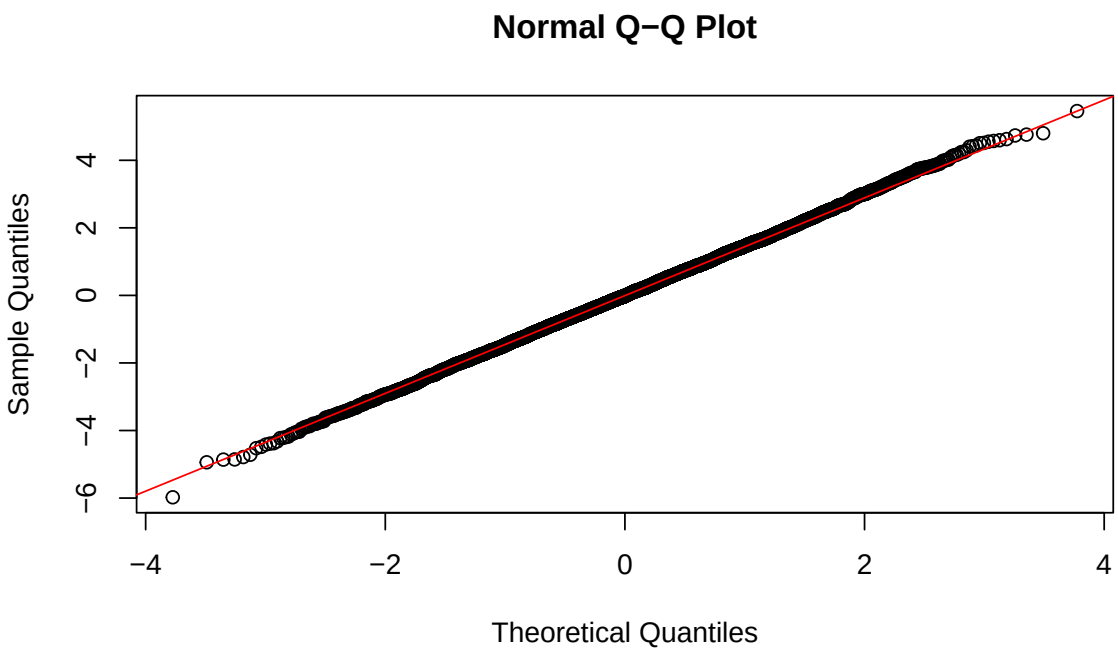
Diagnostics: Evaluating Model Appropriateness

To ensure that the linear mixed-effects model is statistically reliable, several diagnostic checks were performed. These checks help determine whether the model's assumptions are met, and whether the results are reliable.

Normality of Residuals (Q–Q Plot)

The Q–Q plot compares the distribution of the model residuals (errors) to what would be expected if they followed a normal distribution, which is an underlying assumption of linear mixed-effects models.The residuals aligned closely with the expected diagonal pattern, showing only small deviations at the tails.The assumption of normally distributed residuals is largely satisfied. This supports the reliability of the estimated effects.





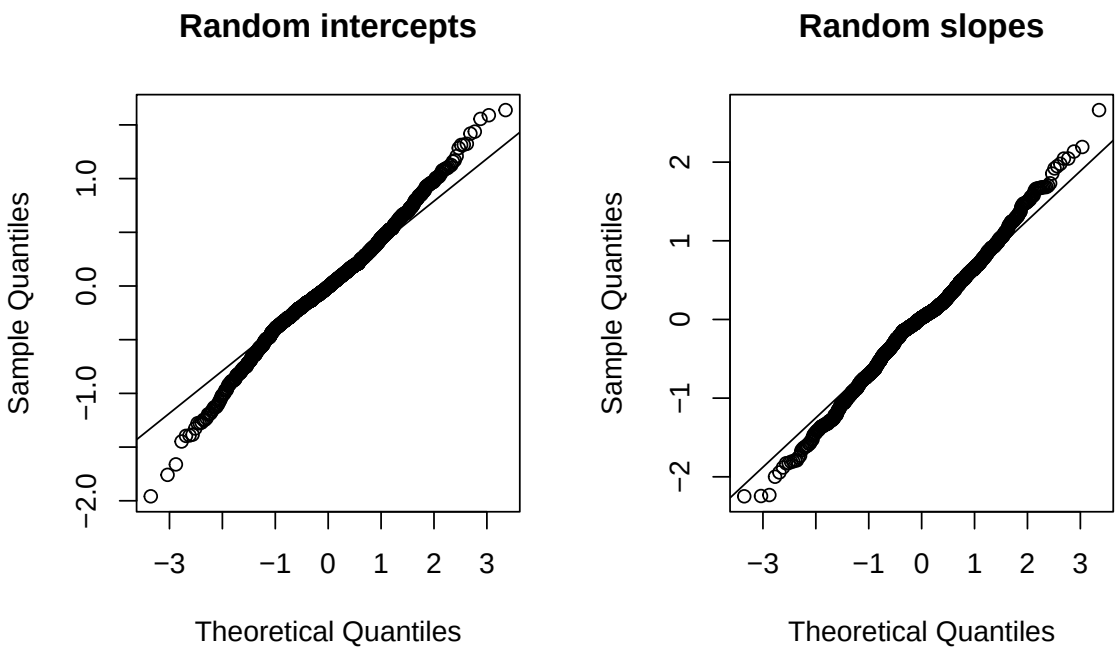
Residuals vs. Fitted Values Plot

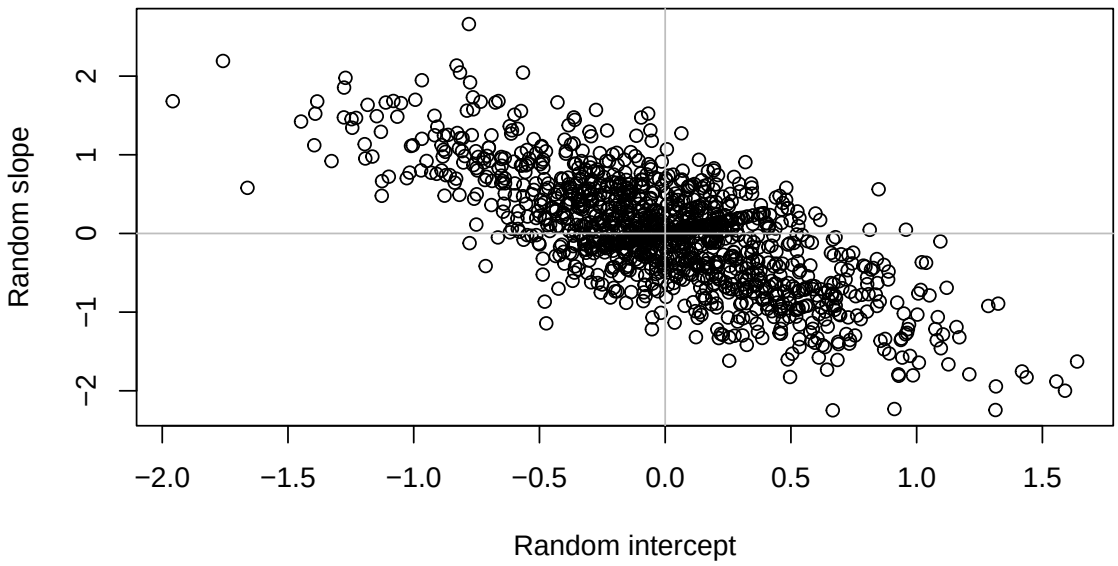
This plot assesses whether residuals are randomly distributed around zero, which reflects the assumption of constant variability ("homoscedasticity").Residuals are spread evenly around zero with no funnel shape or curvature. This indicates Variance is stable across time, The model does not systematically under- or over-predict symptoms and The linearity assumption is appropriate. The model fits the data well and the variability is appropriately captured.

Distribution of Patient-Specific Intercepts and Slopes

To evaluate whether the random-effects model appropriately captured differences between patients, we examined the distribution of the estimated subject-specific intercepts (patients' baseline BPRS scores) and subject-specific slopes (their rates of symptom change over time). The distribution of intercepts showed some variability, indicating that patients entered the study with a wide range of psychiatric severity levels. The slopes also displayed meaningful variation, reflecting that some patients deteriorated faster than others, while a smaller number showed relatively stable symptoms. Both the intercepts and slopes followed approximately normal distributions, with no evidence of heavy tails or extreme outliers. This pattern supports the model assumptions and confirms that the random-effects component is performing as expected accurately capturing differences between patients rather than noise. Overall, the

distributions validate the appropriateness of including random intercepts and random slopes in the model.





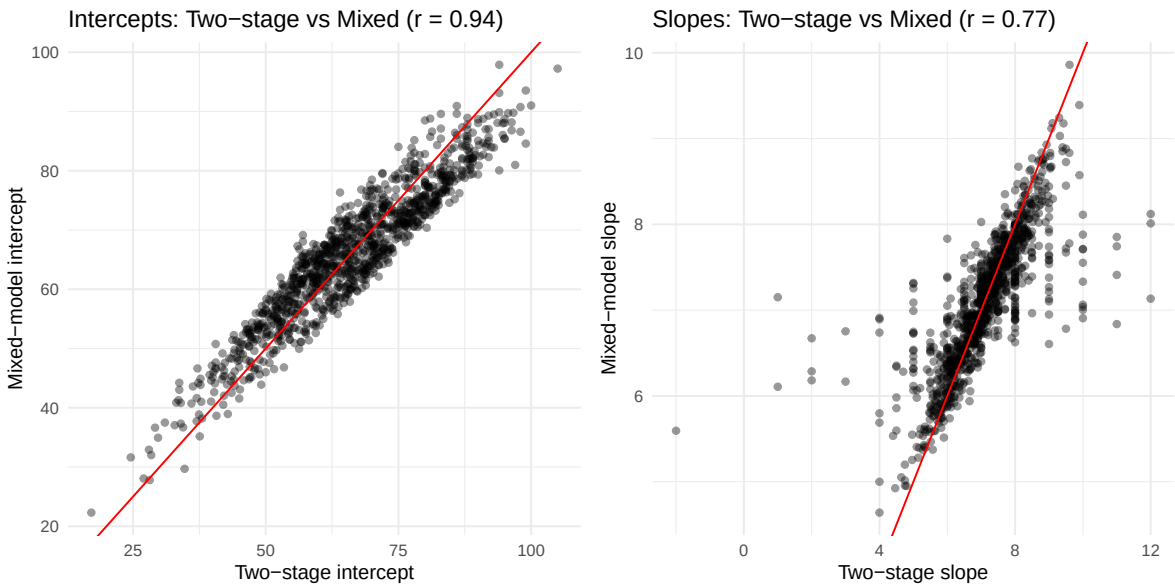
Overall, All diagnostic checks support the validity of the linear mixed-effects model:The residuals behave well (normal distribution and constant variance), The model captures both average disease progression and patient-specific trajectories, The variability across patients justifies including both random intercepts and slopes and The model provides more precise estimates than the two-stage approach and is more statistically efficient.

Comparison With the Two-Stage Analysis

To better understand how each patient's symptoms (BPRS scores) change over time, we estimated individual starting points (intercepts) and rates of change (slopes) in two different ways: 1) Two-stage approach , running a separate linear regression for each patient.(Question 4) 2) Random-effects mixed model using all patients' data together while allowing each patient to have their own intercept and slope.

We then compared the two methods to see whether the more advanced mixed model provides meaningful improvement.

Subject-specific intercepts and slopes from mixed model vs two-stage estimates

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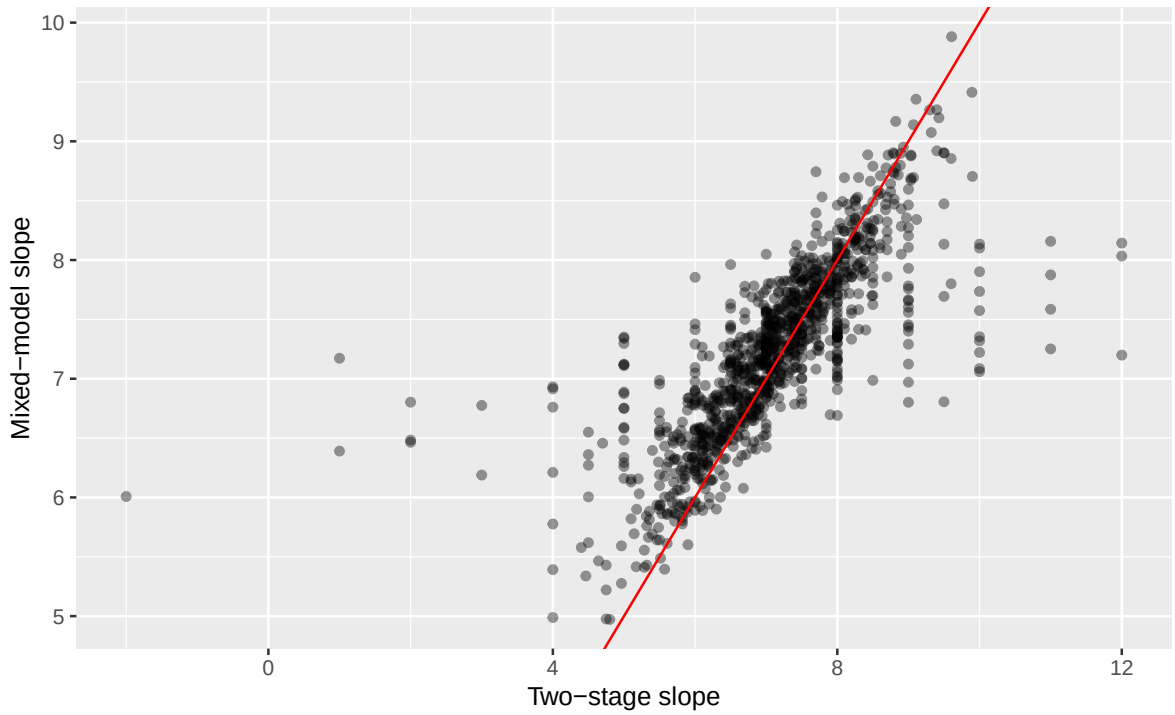

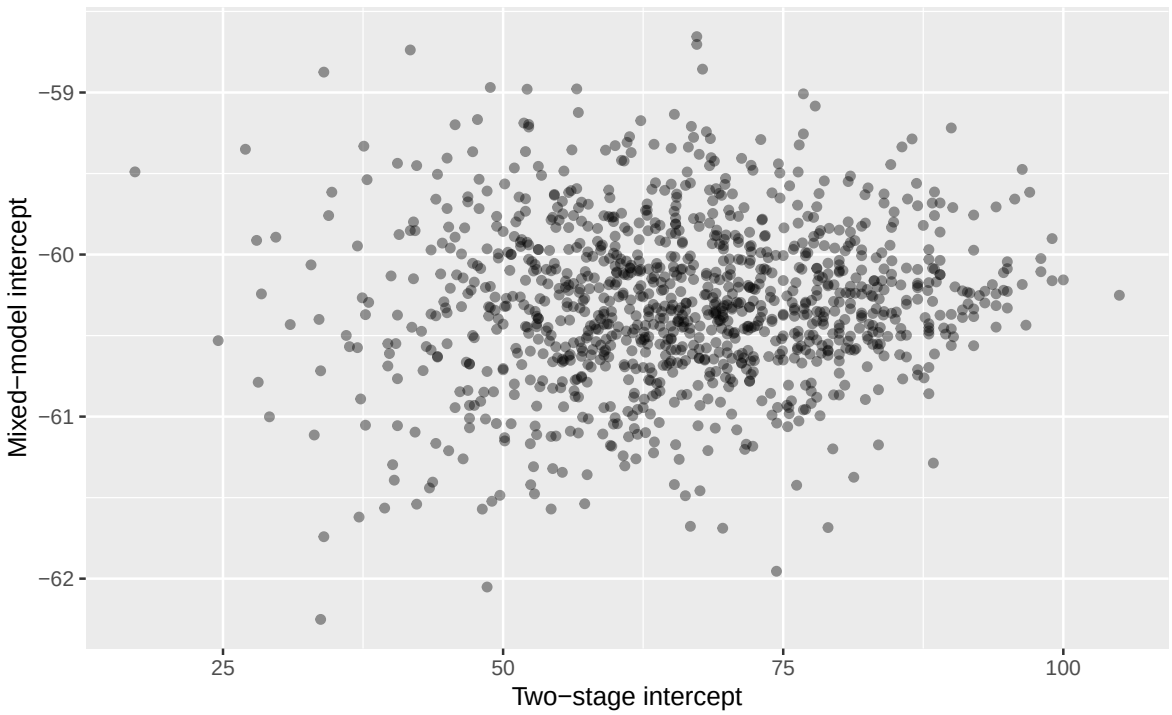

Do both methods give similar patient-specific results?

Yes. When we compared the patient-specific results, we found the two-stage method and the mixed-effects model produced similar patient-specific results. The intercepts (baseline BPRS levels) showed a strong correlation between the two approaches(Figrure 4), indicating that both methods identify the same patients as starting with higher or lower psychiatric severity. Likewise, the individual slopes (rates of symptom change) were strongly positively correlated, demonstrating agreement about which patients deteriorate faster or more slowly over time. This was visually confirmed in the scatterplots (Figure 5), where most points lay close to the 45-degree line of perfect agreement. Together, these findings show that the mixed-effects model and the two-stage analysis yield consistent subject-level estimates, although the mixed model provides them in a more stable and statistically robust way.

Why are the mixed-model estimates still preferred?

Even though the two methods produce similar values, the mixed model is statistically preferred for several reasons:

1) It uses all available data efficiently

The mixed model:Handles patients with different numbers of visits, Corrects for measurement error and Shrinks extreme individual estimates toward the population average (reducing

noise)

2) It adjusts for all predictors The two-stage approach cannot adjust for these characteristics when estimating patient-level slopes. The mixed model does this automatically.
3) It accounts for correlation within patients Longitudinal data contain repeated measures from the same person. The mixed model captures the dependency between these repeated measurements; the two-stage method does not.

Thus, In summary, the random-effects mixed model proved to be the most appropriate and clinically meaningful approach for analyzing individual trajectories in BPRS scores. The model captured substantial and realistic variation between patients in both their baseline psychiatric symptom severity and their rate of change over time. Model diagnostics confirmed that the assumptions of the mixed model were met, with residuals and random effects showing acceptable distributions. When compared to the simpler two-stage method, the mixed model produced highly consistent patient-specific intercepts and slopes, but with greater stability and less sensitivity to noise or sparse data. Furthermore, the mixed model has important advantages: it uses all available information, adjusts individual estimates for relevant clinical predictors, and correctly handles the correlation of repeated measurements within patients. Overall, the random-effects model provides the most robust, reliable, and interpretable description of both population-level trends and individual differences in psychiatric symptom trajectories in this Alzheimer's cohort.

Results

effect	term	estimate	std.error	statistic	p.value
fixed	(Intercept)	-60.29 [-68.48, -52.11]	4.17	-14.46	<0.01
fixed	time	7.22 [7.14, 7.30]	0.04	179.32	<0.01
fixed	sexFemale	-0.16 [-0.33, 0.01]	0.09	-1.84	0.07
fixed	age	1.70 [1.62, 1.78]	0.04	42.36	<0.01
fixed	bmi	0.18 [0.12, 0.23]	0.03	6.54	<0.01
fixed	jobPaid Job	-5.14 [-6.71, -3.56]	0.80	-6.39	<0.01
fixed	adl	0.07 [-0.17, 0.30]	0.12	0.55	0.58
fixed	wzcNursing Home	1.92 [1.72, 2.11]	0.10	19.56	<0.01
fixed	cdrsb0	0.00 [-0.01, 0.02]	0.01	0.26	0.79
fixed	time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.25	<0.01

The linear mixed-effects model was fitted to 6,220 observations from 1,253 patients across 25 trials. The model included random intercepts and slopes for patients and a random intercept for trial. The REML criterion at convergence was 27,121.3, and scaled residuals ranged from -3.64 to 3.32. The patient-level random intercept variance was 0.81, the patient-level random

slope variance was 0.69, and the trial-level intercept variance was 8.00; the residual variance was 2.70. The correlation between patient-specific intercepts and slopes was -0.61.

For the fixed effects, the intercept estimate was -60.29 (SE = 4.17, $p < 0.001$). Time had an estimated coefficient of 7.22 (SE = 0.04, $p < 0.001$). The effect of female sex was -0.16 (SE = 0.09, p = 0.0665), and age had an estimate of 1.70 (SE = 0.04, $p < 0.001$). BMI had an estimated coefficient of 0.18 (SE = 0.03, $p < 0.001$), while being in paid employment had a coefficient of -5.14 (SE = 0.80, $p < 0.001$). The ADL score estimate was 0.07 (SE = 0.12, p = 0.5791). Nursing home residence had an estimate of 1.92 (SE = 0.10, $p < 0.001$). Baseline CDR-SB had an estimate of 0.002 (SE = 0.008, p = 0.7924). The interaction between time and baseline CDR-SB was -0.022 (SE = 0.004, $p < 0.001$).

Summary Conclusion to a clinician

Using a mixed-effects model, we examined how psychiatric symptoms change over time in patients with Alzheimer's disease. This modelling approach accounts for individual differences in both baseline symptom severity and rates of progression.

The analysis showed that psychiatric symptoms worsen steadily over time, with patients on average increasing by about 7 BPRS points per year. However, not all patients start at the same level or decline at the same speed. Our model captured this variation and revealed important patient characteristics that help explain these differences.

Patients who were older, had higher BMI, poorer daily functioning (ADL), lived in a nursing home, or did not have a paid job tended to have more severe psychiatric symptoms at baseline. In addition, patients with more impaired cognition at diagnosis (higher CDR-SB) showed faster worsening of psychiatric symptoms over time. This directly answers our main question at the population level: BPRS worsens over time, and the speed of worsening is associated with baseline cognitive status.

For the biomarkers, we focus on ABPET0 and CDR-SB. ABPET0 is retained in the marginal model, but its coefficient is small and the confidence interval is wide, so its association with BPRS is uncertain. The LOESS plot of BPRS versus ABPET0 by visit year shows an unstable curved pattern, with a peak at year 3 that is not repeated at other visits. This could be due to data missingness. TAUPET0, on the other hand, was dropped because from year 3 onwards it has no variability, so it cannot explain differences in BPRS.

The covariance part of the model lets each visit have its own variability and allows all time points within a patient to be correlated without imposing a correlation structure. This matches what we saw in the exploratory plots repeated BPRS measurements from the same patient are strongly correlated, especially when visits are close in time, and the spread increases

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Finally, this marginal model has some limitations compared with a mixed-effects model. It provides population-average effects but not patient-specific trajectories, and it does not explicitly model differences between clinical centres. The LOESS plot by centre shows centres with roughly parallel but clearly non overlapping curves for majority of centres, suggesting centre level differences in BPRS levels over time. In the current model, this extra variation is absorbed into the residual covariance rather than being separated out. In Question 5 we therefore move to a mixed-effects model with random intercepts and slopes for patients, and a random intercept for trial, to better describe individual and centre-specific disease trajectories while still addressing the main objective of how BPRS changes over time, and how those changes relate to baseline clinical characteristics, cognition, and biomarkers.

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In the first stage of a two-stage analysis, we fit a separate linear regression model for each patient to estimate their individual intercept and slope over time. For each patient i, the model is:

$$Y_{ij} = \beta_{0i} + \beta_{1i} \cdot \text{time}_j + e_{ij}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
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- β_{1i} is the patient-specific slope (estimated rate of change in BPRS).
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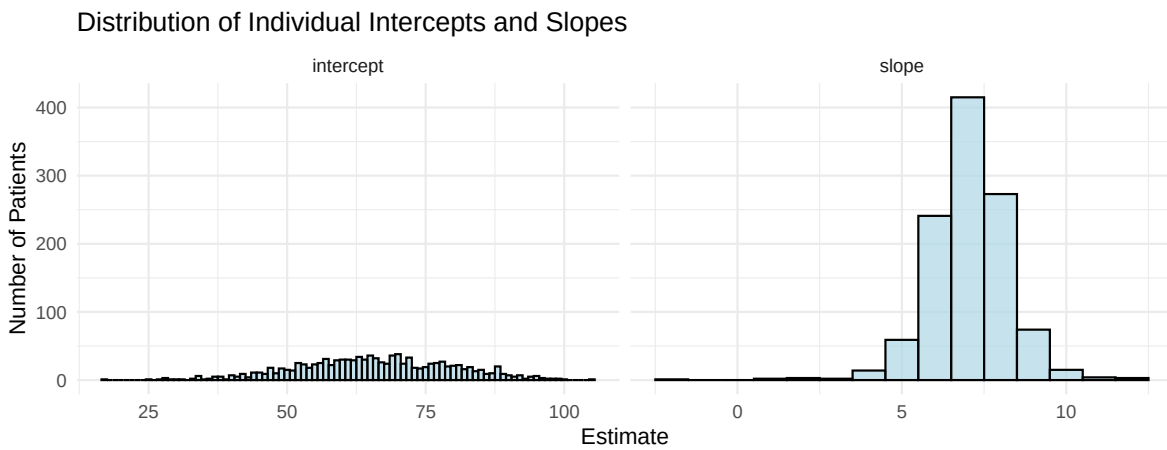

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Model for the Intercepts

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$$\begin{aligned}\hat{\beta}_{0i} =& \gamma_{00} + \gamma_{01} \cdot \text{sex}_i + \gamma_{02} \cdot \text{age}_i + \gamma_{03} \cdot \text{bmi}_i + \gamma_{04} \cdot \text{job}_i + \gamma_{05} \cdot \text{adl}_i \\ &+ \gamma_{06} \cdot \text{wzc}_i + \gamma_{07} \cdot \text{cdrsb0}_i + \gamma_{08} \cdot \text{abpet0}_i + \gamma_{09} \cdot \text{taupet0}_i + \epsilon_{0i}\end{aligned}$$

where:

- $\hat{\beta}_{0i}$ is the estimated intercept for patient i from Stage 1.
- γ_{00} is the average intercept for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{01} is the intercept difference for Female vs Male.
- γ_{02} is the intercept change per 1-year increase in age.
- γ_{03} is the intercept change per 1-unit increase in BMI.
- γ_{04} is the intercept difference for Paid Job vs No Paid Job.
- γ_{05} is the intercept change per 1-unit increase in ADL.
- γ_{06} is the intercept difference for Nursing Home vs Lives at Home.
- γ_{07} is the intercept change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{08} is the intercept change per 1-unit increase in baseline Amyloid PET (abpet0).

- γ_{09} is the intercept change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{0i} is the error term.

Table 10: Two-Stage Intercept Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-130.85 [-137.31, -124.39]	3.29	-39.76	<0.01
sexFemale	-0.22 [-0.51, 0.06]	0.15	-1.52	0.13
age	2.36 [2.32, 2.41]	0.02	101.88	<0.01
bmi	0.51 [0.45, 0.58]	0.03	15.07	<0.01
jobPaid Job	-18.28 [-19.25, -17.32]	0.49	-37.15	<0.01
adl	2.03 [1.90, 2.15]	0.06	31.98	<0.01
wzcNursing Home	1.54 [1.22, 1.86]	0.16	9.41	<0.01
cdrsb0	0.00 [-0.02, 0.02]	0.01	0.12	0.90
abpet0	0.35 [-0.16, 0.86]	0.26	1.35	0.18
taupet0	0.53 [-2.27, 3.33]	1.43	0.37	0.71

Model for the Slopes

Similarly, we model the estimated patient-specific slopes ($\hat{\beta}_{1i}$) as a function of baseline covariates to see what predicts the rate of change in BPRS. The model is:

$$\begin{aligned}\hat{\beta}_{1i} =& \gamma_{10} + \gamma_{11} \cdot \text{sex}_i + \gamma_{12} \cdot \text{age}_i + \gamma_{13} \cdot \text{bmi}_i + \gamma_{14} \cdot \text{job}_i + \gamma_{15} \cdot \text{adl}_i \\ &+ \gamma_{16} \cdot \text{wzc}_i + \gamma_{17} \cdot \text{cdrsb0}_i + \gamma_{18} \cdot \text{abpet0}_i + \gamma_{19} \cdot \text{taupet0}_i + \epsilon_{1i}\end{aligned}$$

where:

- $\hat{\beta}_{1i}$ is the estimated slope for patient i from Stage 1.
- γ_{10} is the average slope for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{11} is the slope difference for Female vs Male.
- γ_{12} is the slope change per 1-year increase in age.
- γ_{13} is the slope change per 1-unit increase in BMI.
- γ_{14} is the slope difference for Paid Job vs No Paid Job.
- γ_{15} is the slope change per 1-unit increase in ADL.
- γ_{16} is the slope difference for Nursing Home vs Lives at Home.
- γ_{17} is the slope change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{18} is the slope change per 1-unit increase in baseline Amyloid PET (abpet0).

- γ_{19} is the slope change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{1i} is the error term.

Table 11: Two-Stage Slope Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	6.37 [2.97, 9.77]	1.73	3.68	<0.01
sexFemale	-0.04 [-0.19, 0.11]	0.08	-0.55	0.58
age	0.01 [-0.01, 0.04]	0.01	1.02	0.31
bmi	0.00 [-0.03, 0.04]	0.02	0.09	0.93
jobPaid Job	-0.34 [-0.85, 0.17]	0.26	-1.32	0.19
adl	0.05 [-0.02, 0.11]	0.03	1.37	0.17
wzcNursing Home	0.19 [0.02, 0.36]	0.09	2.23	0.03
cdrsb0	-0.02 [-0.03, -0.01]	0.01	-4.30	<0.01
abpet0	-0.07 [-0.34, 0.20]	0.14	-0.53	0.60
taupet0	-0.13 [-1.61, 1.34]	0.75	-0.17	0.86

Question 5

Plausible random effects model

To understand how psychiatric symptoms evolve over time in Alzheimer's disease, we developed a statistical model that accounts for repeated measurements in each patient. This model allows each patient to have their own starting level of symptoms and their own rate of change over time. The outcome of interest was the Brief Psychiatric Rating Scale (BPRS), measured annually over six years.

We model the longitudinal BPRS scores using a linear mixed-effects model with random effects at both the patient and trial level:

$$\text{BPRS}_{it} = (\beta_0+b_{0i}+u_{0j})+(\beta_1+b_{1i})\,\text{time}_{it}+\beta_2\text{sex}_i+\beta_3\text{age}_i+\beta_4\text{bmi}_i+\beta_5\text{job}_i+\beta_6\text{adl}_i+\beta_7\text{wzc}_i+\beta_8\text{cdrsb0}_i+\beta_9(\text{time}_i$$

where:

- (BPRS_{it}) is the BPRS score for patient (i) in trial (j) at time (t)
- (_0, ..., _9) are **fixed (population-level) effects**

- (b_{0i}) is the **patient-specific random intercept**
- (b_{1i}) is the **patient-specific random slope** for time
- (u_{0j}) is the **trial-specific random intercept**
- (_{it}) is the **residual error term**

The patient-level random effects are assumed to follow a bivariate normal distribution:

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{b0}^2 & \sigma_{b0,b1} \\ \sigma_{b0,b1} & \sigma_{b1}^2 \end{pmatrix} \right),$$

and the trial-level random intercept follows:

$$u_{0j} \sim N(0, \sigma_{\text{trial}}^2).$$

Model Motivation and Justification

Here, We chose to fit linear mixed effects model because the dataset contains repeated BPRS measurements for the same patients over 6 years and across multiple clinical trials, with clear variability both in baseline symptom severity and rate of change between patients. This model was appropriate because it:

1) Explicitly accounts for correlation within patients (repeated measures from the same individual),
2) Allows patient-specific trajectories (subject-specific intercepts and slopes),
3) Uses all available data, including patients with incomplete follow-up.
4) and Adjusts the individual trajectories for baseline patient characteristics, improving precision.

Choice of Technique

An unstructured covariance matrix was used for the random effects, allowing the intercept and slope to be correlated. The model included:

1) Random intercepts and random slopes for each patient, reflecting that individuals differ both in their baseline psychiatric symptom severity and in their rate of change over time.

2) A random intercept for trial was added to account for systematic differences across the 25 clinical trials.
3) Fixed effects included all clinically relevant baseline covariates (age, sex, BMI, job status, ADL, living situation, CDR-SB, amyloid PET), ensuring adjustment for major patient characteristics.
4) An interaction term between time $\times$ CDRSB was included to assess whether cognitive impairment at baseline affects the rate of psychiatric symptom progression.

Estimation Method

The model was fitted using Restricted Maximum Likelihood (REML), which provides unbiased and stable estimates of variance components, especially because our dataset had many clusters (patients, trials).We also preferred this (REML)for comparing models that differ in their random-effects structure.

Assumptions made

1) Linearity: BPRS changes linearly over the follow-up period.
2) Normality: Random effects and residual errors are approximately normally distributed.
3) Independence: After accounting for random effects, residuals are independent.
4) Missing data (dropout) are at least missing at random (MAR) conditional on the covariates and random effects.
5) Correct Covariance Structure: The random intercept–slope covariance captures individual variability adequately

Advantages

1) Properly accounts for correlation between repeated measures within patients.
2) Handles unbalanced data, allowing inclusion of patients with incomplete follow-up.
3) Produces subject-specific predictions (intercepts and slopes).
4) The model incorporates trial-level variability, improving generalizability.
5) More efficient and less biased than two-stage or marginal models for this scientific question.

Disadvantages / Potential Problems

1) More complex to interpret for non-statisticians.
2) Results depend on correct specification of random-effects structure.
3) Convergence issues can arise with complex models.

4) Assumptions such as normality of random effects and linearity may not fully hold in real data.

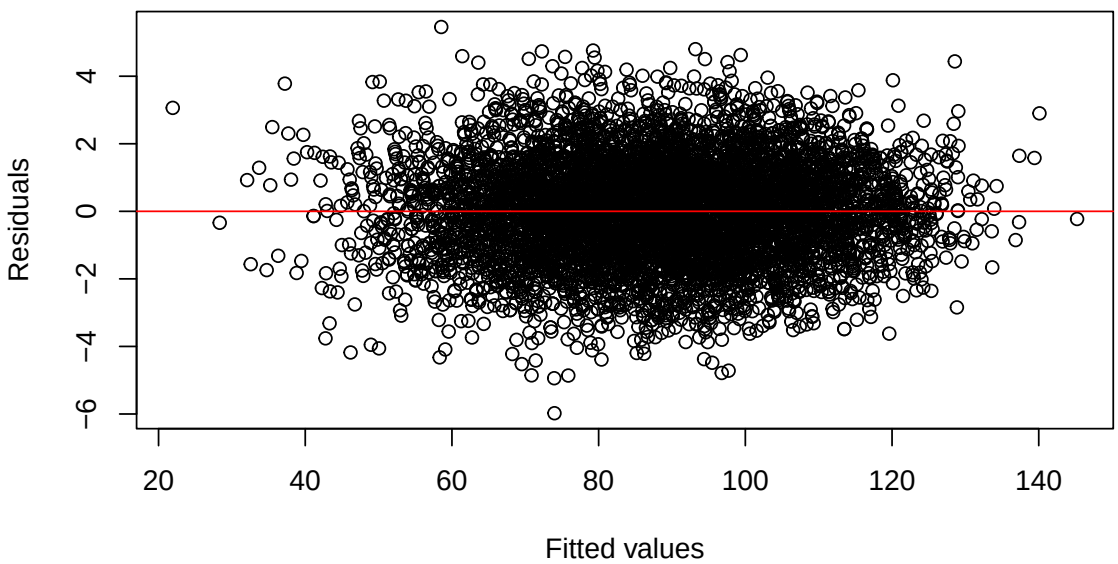
5) MAR assumption cannot be fully verified.

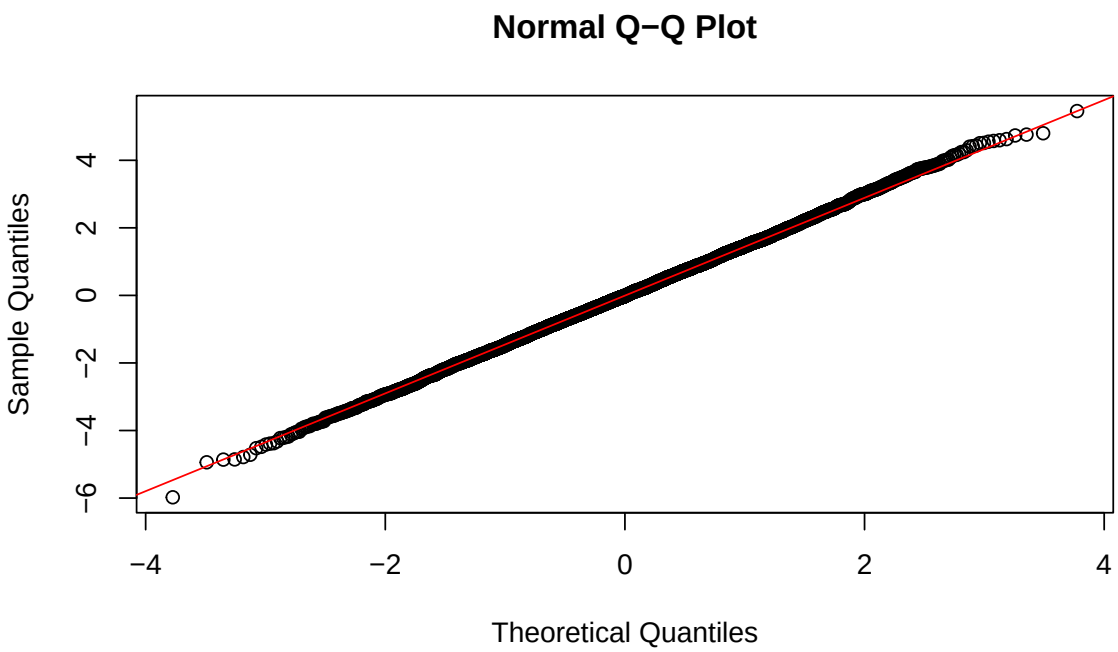
Diagnostics: Evaluating Model Appropriateness

To ensure that the linear mixed-effects model is statistically reliable, several diagnostic checks were performed. These checks help determine whether the model's assumptions are met, and whether the results are reliable.

Normality of Residuals (Q–Q Plot)

The Q–Q plot compares the distribution of the model residuals (errors) to what would be expected if they followed a normal distribution, which is an underlying assumption of linear mixed-effects models.The residuals aligned closely with the expected diagonal pattern, showing only small deviations at the tails.The assumption of normally distributed residuals is largely satisfied. This supports the reliability of the estimated effects.





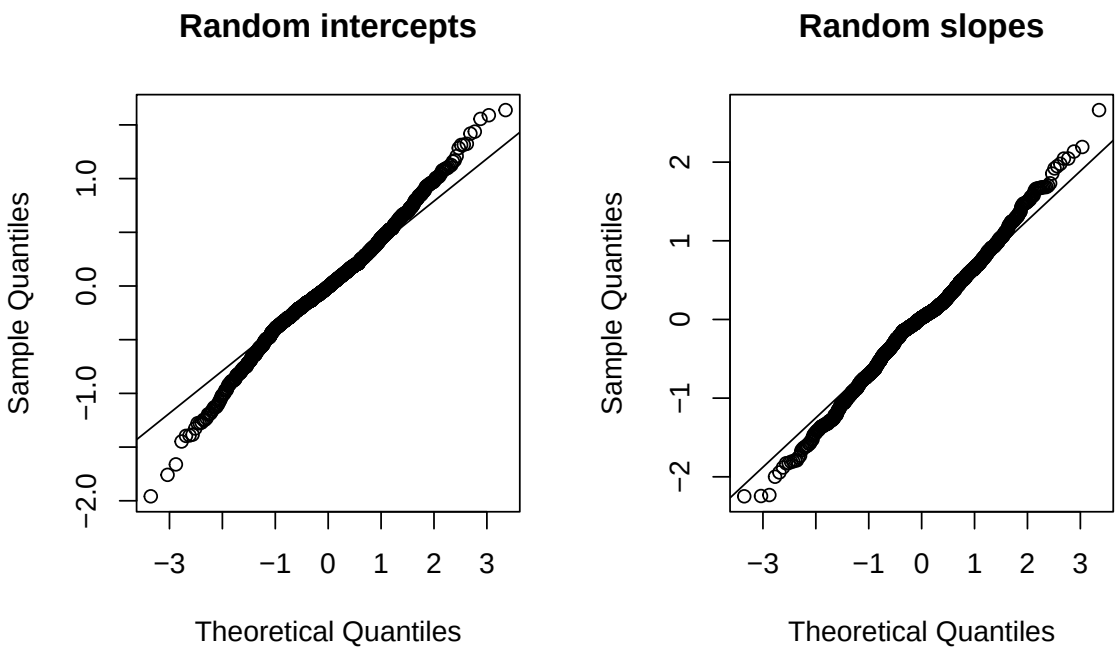
Residuals vs. Fitted Values Plot

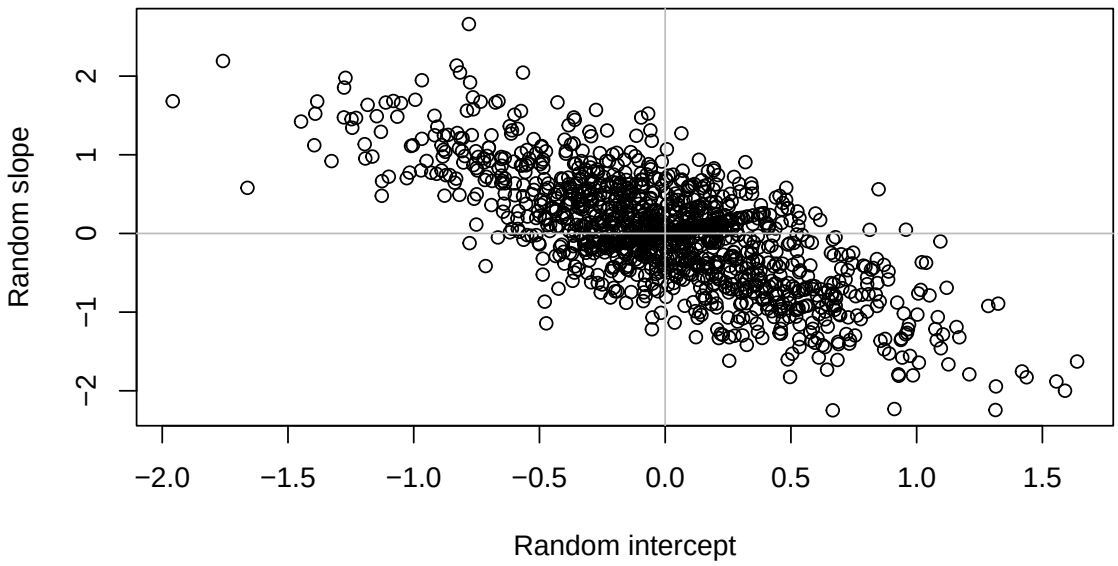
This plot assesses whether residuals are randomly distributed around zero, which reflects the assumption of constant variability ("homoscedasticity").Residuals are spread evenly around zero with no funnel shape or curvature. This indicates Variance is stable across time, The model does not systematically under- or over-predict symptoms and The linearity assumption is appropriate. The model fits the data well and the variability is appropriately captured.

Distribution of Patient-Specific Intercepts and Slopes

To evaluate whether the random-effects model appropriately captured differences between patients, we examined the distribution of the estimated subject-specific intercepts (patients' baseline BPRS scores) and subject-specific slopes (their rates of symptom change over time). The distribution of intercepts showed some variability, indicating that patients entered the study with a wide range of psychiatric severity levels. The slopes also displayed meaningful variation, reflecting that some patients deteriorated faster than others, while a smaller number showed relatively stable symptoms. Both the intercepts and slopes followed approximately normal distributions, with no evidence of heavy tails or extreme outliers. This pattern supports the model assumptions and confirms that the random-effects component is performing as expected accurately capturing differences between patients rather than noise. Overall, the

distributions validate the appropriateness of including random intercepts and random slopes in the model.





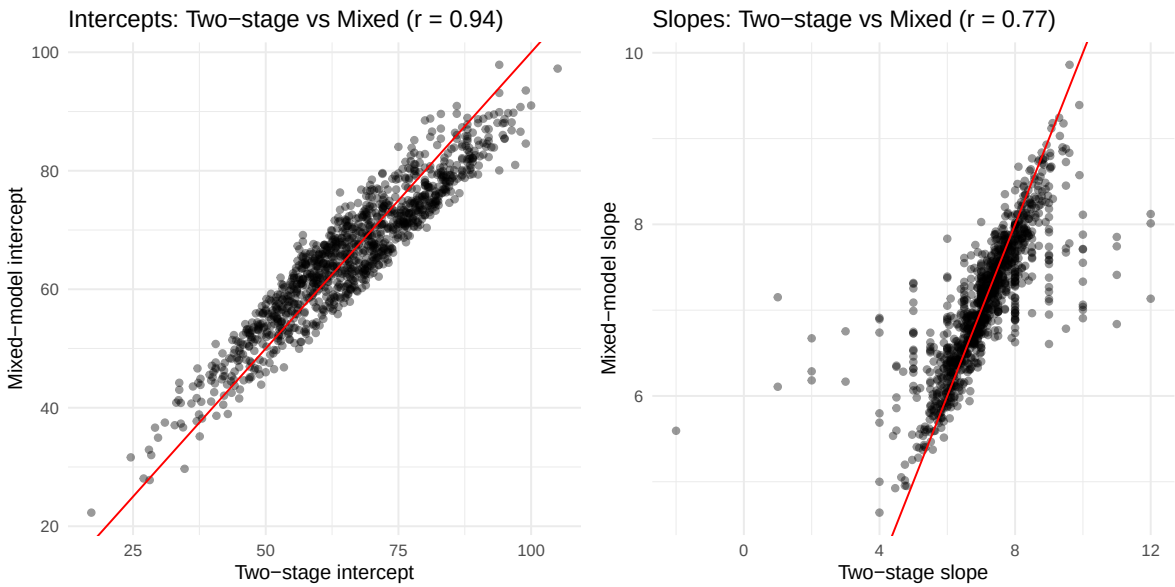
Overall, All diagnostic checks support the validity of the linear mixed-effects model:The residuals behave well (normal distribution and constant variance), The model captures both average disease progression and patient-specific trajectories, The variability across patients justifies including both random intercepts and slopes and The model provides more precise estimates than the two-stage approach and is more statistically efficient.

Comparison With the Two-Stage Analysis

To better understand how each patient's symptoms (BPRS scores) change over time, we estimated individual starting points (intercepts) and rates of change (slopes) in two different ways: 1) Two-stage approach , running a separate linear regression for each patient.(Question 4) 2) Random-effects mixed model using all patients' data together while allowing each patient to have their own intercept and slope.

We then compared the two methods to see whether the more advanced mixed model provides meaningful improvement.

Subject-specific intercepts and slopes from mixed model vs two-stage estimates

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[1] 0.07416293
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[1] 0.762492
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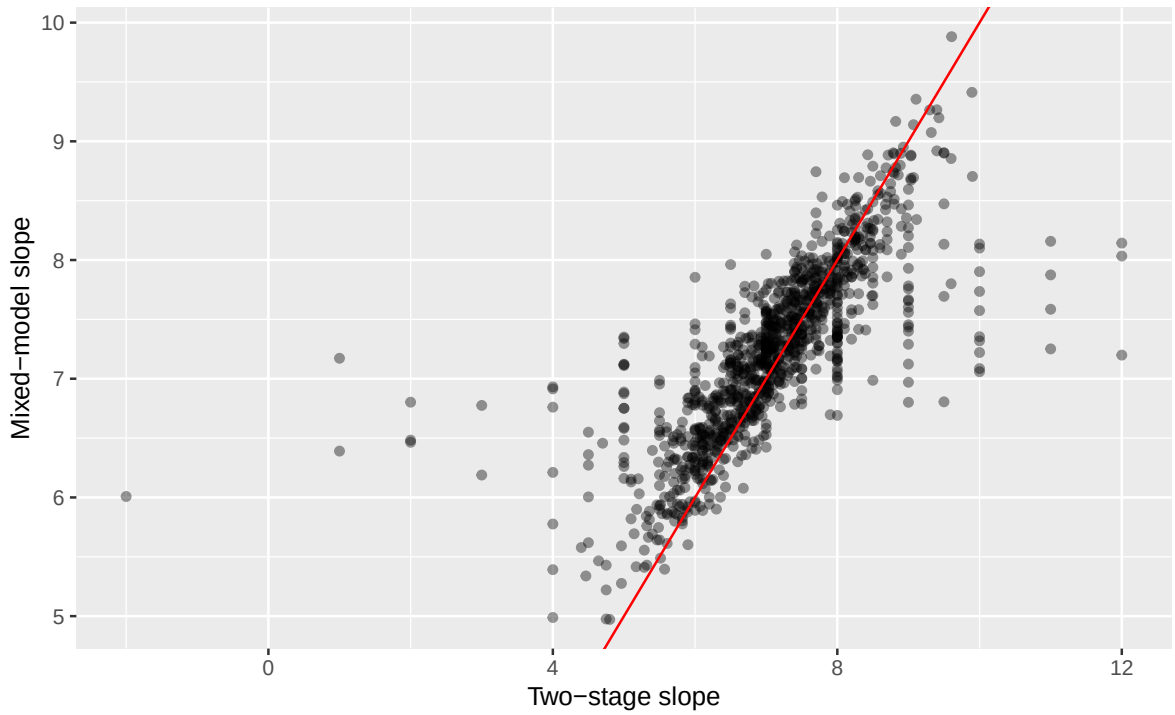

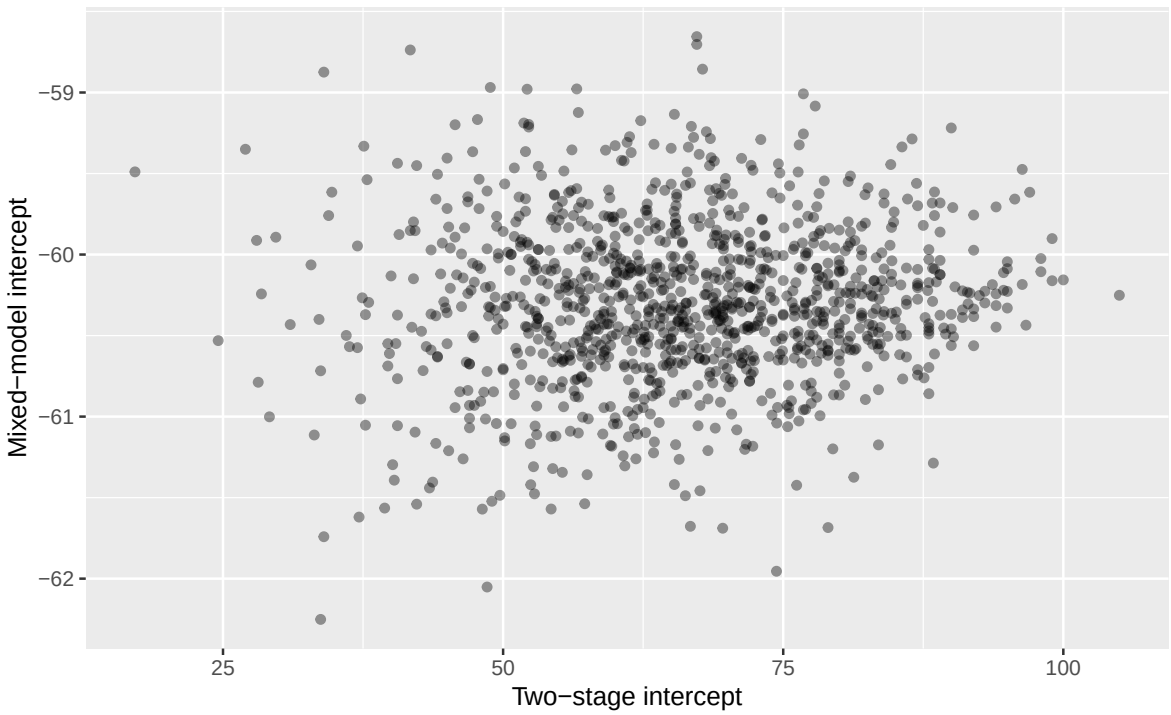

Do both methods give similar patient-specific results?

Yes. When we compared the patient-specific results, we found the two-stage method and the mixed-effects model produced similar patient-specific results. The intercepts (baseline BPRS levels) showed a strong correlation between the two approaches(Figrure 4), indicating that both methods identify the same patients as starting with higher or lower psychiatric severity. Likewise, the individual slopes (rates of symptom change) were strongly positively correlated, demonstrating agreement about which patients deteriorate faster or more slowly over time. This was visually confirmed in the scatterplots (Figure 5), where most points lay close to the 45-degree line of perfect agreement. Together, these findings show that the mixed-effects model and the two-stage analysis yield consistent subject-level estimates, although the mixed model provides them in a more stable and statistically robust way.

Why are the mixed-model estimates still preferred?

Even though the two methods produce similar values, the mixed model is statistically preferred for several reasons:

1) It uses all available data efficiently

The mixed model:Handles patients with different numbers of visits, Corrects for measurement error and Shrinks extreme individual estimates toward the population average (reducing

noise)

2) It adjusts for all predictors The two-stage approach cannot adjust for these characteristics when estimating patient-level slopes. The mixed model does this automatically.
3) It accounts for correlation within patients Longitudinal data contain repeated measures from the same person. The mixed model captures the dependency between these repeated measurements; the two-stage method does not.

Thus, In summary, the random-effects mixed model proved to be the most appropriate and clinically meaningful approach for analyzing individual trajectories in BPRS scores. The model captured substantial and realistic variation between patients in both their baseline psychiatric severity and their rate of symptom progression. Model diagnostics confirmed that the assumptions of the mixed model were met, with residuals and random effects showing acceptable distributions. When compared to the simpler two-stage method, the mixed model produced highly consistent patient-specific intercepts and slopes, but with greater stability and less sensitivity to noise or sparse data. Furthermore, the mixed model has important advantages: it uses all available information, adjusts individual estimates for relevant clinical predictors, and correctly handles the correlation of repeated measurements within patients. Overall, the random-effects model provides the most robust, reliable, and interpretable description of both population-level trends and individual differences in psychiatric symptom trajectories in this Alzheimer's cohort.

Results

effect	term	estimate	std.error	statistic	p.value
fixed	(Intercept)	-60.29 [-68.48, -52.11]	4.17	-14.46	<0.01
fixed	time	7.22 [7.14, 7.30]	0.04	179.32	<0.01
fixed	sexFemale	-0.16 [-0.33, 0.01]	0.09	-1.84	0.07
fixed	age	1.70 [1.62, 1.78]	0.04	42.36	<0.01
fixed	bmi	0.18 [0.12, 0.23]	0.03	6.54	<0.01
fixed	jobPaid Job	-5.14 [-6.71, -3.56]	0.80	-6.39	<0.01
fixed	adl	0.07 [-0.17, 0.30]	0.12	0.55	0.58
fixed	wzcNursing Home	1.92 [1.72, 2.11]	0.10	19.56	<0.01
fixed	cdrsb0	0.00 [-0.01, 0.02]	0.01	0.26	0.79
fixed	time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.25	<0.01

The linear mixed-effects model was fitted to 6,220 observations from 1,253 patients across 25 trials. The model included random intercepts and slopes for patients and a random intercept for trial. The REML criterion at convergence was 27,121.3, and scaled residuals ranged from -3.64 to 3.32. The patient-level random intercept variance was 0.81, the patient-level random

slope variance was 0.69, and the trial-level intercept variance was 8.00; the residual variance was 2.70. The correlation between patient-specific intercepts and slopes was -0.61.

For the fixed effects, the intercept estimate was -60.29 (SE = 4.17, $p < 0.001$). Time had an estimated coefficient of 7.22 (SE = 0.04, $p < 0.001$). The effect of female sex was -0.16 (SE = 0.09, $p = 0.0665$), and age had an estimate of 1.70 (SE = 0.04, $p < 0.001$). BMI had an estimated coefficient of 0.18 (SE = 0.03, $p < 0.001$), while being in paid employment had a coefficient of -5.14 (SE = 0.80, $p < 0.001$). The ADL score estimate was 0.07 (SE = 0.12, $p = 0.5791$). Nursing home residence had an estimate of 1.92 (SE = 0.10, $p < 0.001$). Baseline CDRSB had an estimate of 0.002 (SE = 0.008, $p = 0.7924$). The interaction between time and baseline CDRSB was -0.022 (SE = 0.004, $p < 0.001$).

Summary Conclusion to a clinician

Using a mixed-effects model, we examined how psychiatric symptoms change over time in patients with Alzheimer's disease. This modelling approach accounts for individual differences in both baseline symptom severity and rates of progression.

The analysis showed that psychiatric symptoms worsen steadily over time, with patients on average increasing by about 7 BPRS points per year. However, not all patients start at the same level or decline at the same speed. Our model captured this variation and revealed important patient characteristics that help explain these differences.

Patients who were older, had higher BMI, poorer daily functioning (ADL), lived in a nursing home, or did not have a paid job tended to have more severe psychiatric symptoms at baseline. In addition, patients with more impaired cognition at diagnosis (higher CDR-SB) showed faster worsening of psychiatric symptoms over time. PET biomarkers for amyloid did not significantly improve prediction beyond these clinical measures.

Overall, this approach provides a detailed picture of how psychiatric symptoms evolve in Alzheimer's disease. It shows which baseline factors are linked with faster decline, and it gives individualized estimates of symptom progression. These findings can help to better understand patient variability, identify patients who may be at higher risk for rapid deterioration, and guide monitoring and care strategies over the course of the disease.

Reflection on parameterization

The chosen parameterization had two main components:

Mean structure (fixed effects): We specified a linear effect of time, adjusted for sex, age, BMI, job status, ADL, nursing home residence (WZC), baseline CDR-SB, tau PET, amyloid PET, and the interaction between time and CDR-SB. This reflects a clinical hypothesis that:

1) symptom severity depends on demographic, functional, and disease severity markers at baseline, and

2) cognitive status modifies the trajectory of psychiatric symptoms (time $\times$ CDRSB0).

A key simplification is assuming a linear trajectory in time; in reality, progression could be non-linear (e.g., faster early or late). If time effects are non-linear, a model with splines or time as a factor could fit better, but at the cost of interpretability and more parameters.

Covariance structure (random effects): We chose random intercepts and random slopes for time at the patient level, with an unstructured 2×2 covariance matrix. This is a very interpretable parameterization: one parameter for variability in baseline severity, one for variability in slope, and one for their correlation. Compared with fully unstructured residual covariance (as in the GLS model), this is more parsimonious and more directly linked to clinical ideas like "patients start differently" and "patients change at different speeds".

The model therefore strikes a balance between flexibility (allowing heterogeneity in trajectories) and interpretability (simple linear time, clinically meaningful random effects). In future, we might consider testing alternative parameterizations (e.g., non-linear time, random effects on additional covariates) but for this dataset, the chosen structure appears adequate and clinically interpretable.

Recommendations for future similar experiments

From a modeling and design perspective, several recommendations emerge:

1) systematic recording of dropout reasons: Since dropout can bias longitudinal analyses, future studies should routinely document why patients leave the study (death, institutionalization, refusal, health decline). This would allow sensitivity analyses under different missing-data assumptions (MAR vs MNAR).

2) Richer covariate information on treatment and care: Including medication use (e.g., antipsychotics, antidepressants), psychosocial interventions, caregiver burden, and comorbidities might help explain additional between-patient variation and could identify modifiable risk factors for rapid deterioration.

3) Parallel assessment of multiple outcome domains Since cognition, function, and psychiatric symptoms are closely related in Alzheimer's disease, future experiments could be designed to jointly model these domains rather than analyzing each in isolation. This would provide a more integrated picture of disease progression and may better inform clinical decision-making.

Variance components from mixed model

Table 12: Random effects: variance components (lmer)

	Groups	Name	Variance	Std.Dev.	Corr
(Intercept)	patid	(Intercept)	0.807	0.898	NA
time	patid	time	0.686	0.828	-0.611
(Intercept)1	trial	(Intercept)	8.004	2.829	NA
1	Residual		2.698	1.643	NA